



An Introduction to AI & where Computer Vision Meets Creation

6 October
2025

v. 2.1



Course Goals

Intro to AICV

Comprehensive understanding of AI and Computer Vision and its role in the real-world applications

PixelMachineLab

Hands-on experience with Python and familiarization with relevant tools and libraries

Hackathon + Project

Apply innovative methods like Future Thinking, Garage Frameworks modern technologies for a group project in the context of Computer Vision.

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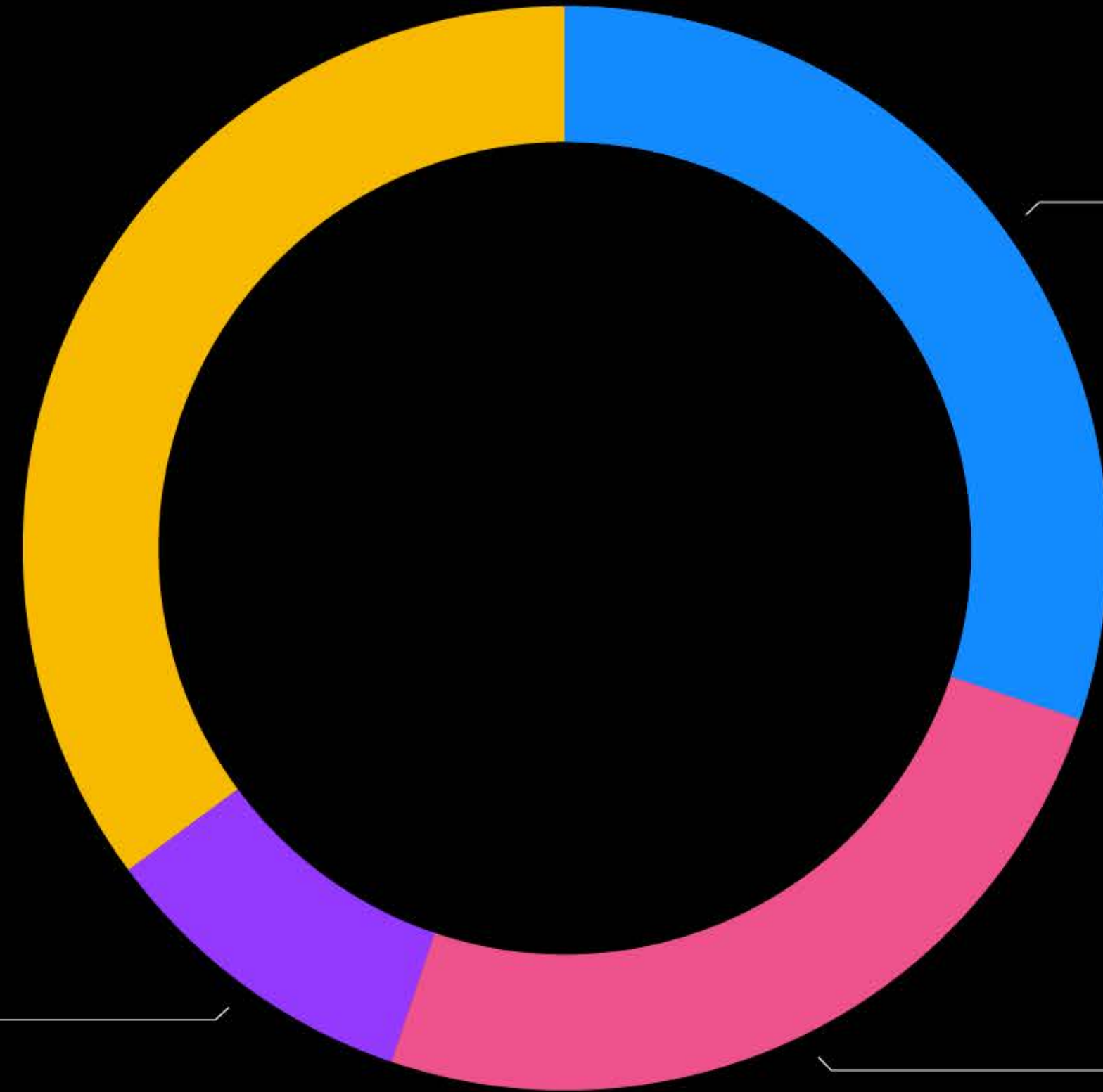




Activity Distribution — AI & Computer Vision (5 ECTS)

- Theory
- Labs
- Micro Deep Dives
- Hackathon & Group Project

Hackathon &
Group Project
25%



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Seminar mixed with theory & practical sessions

Theory Journey – How the course builds conceptually

From Vision to Intelligence:

How machines learn to see

Recognition & Tracking:

Teaching machines to follow what they see

Deep Learning for Vision:

Neural networks that understand images

Generative & Responsible Vision:

When machines start to create — and deceive

Applied & Creative Vision: **Designing with AI & looking ahead**

The Practical Journey – Creating the Pixel Machine Lab

Image Manipulation with OpenCV · Mini Image Gallery

Face & Object Tracking · Dataset Annotation

Training Custom Models · Gesture Recognition · YOLOv9

Micro Deep Dives:
Generative Vision · Diffusion · Ethics · Bias · AI Tools

Spatial AI and 3D Point Cloud Exploration with Open3D

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Micro Deep Dives (November)

👥 Working **pairs** on advanced Computer Vision topics
→ from **Generative AI** to **Ethics and Bias**

🕒 Replace ~8h of regular lectures
→ focus on **independent, research-driven learning**

🧩 Not formally graded
→ but strong performance can **positively influence your final grade**

💡 Goal: connect theory, experimentation, and reflection
→ bring fresh insights into our shared learning journey

Group Project (Jan/Feb)

👥 Work in teams of 3–5 students
→ collaborate on a Computer Vision project of your choice

🧠 Apply everything learned throughout the course
→ from **detection and tracking** to **generative and spatial AI**

📅 Present your project in **February**
→ count toward your **final grade (5 ECTS workload)**

🎯 Projects are **graded individually**
→ based on contribution, understanding, and presentation

💡 Topics are ideated during the Future Thinking Workshop & Kick Off in Hackathon in January





HowWeUsePythoninThisCourse

Python is not the subject—it's our
tool to explore vision and
intelligence.

We use Python as a tool, not a goal:

- No coding background required
- We use it to understand *how* vision systems work

Focus on Logic, not Syntax

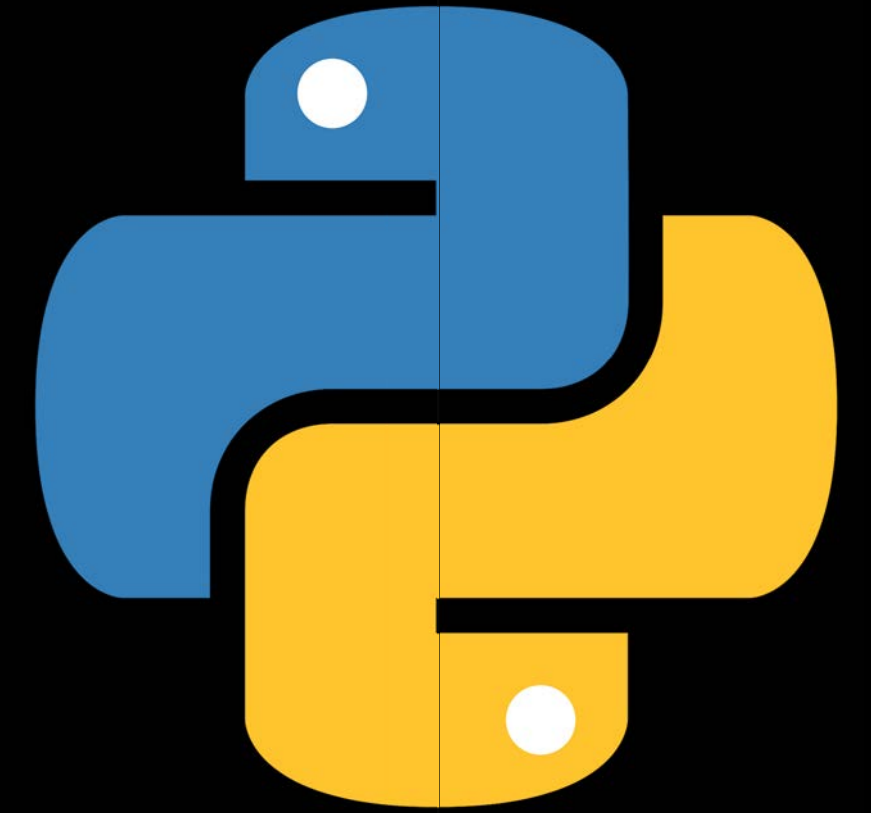
- Work with prepared templates
- Change, test, and observe results visually

AI-Assisted Coding = Allowed (with Responsibility)

- Use ChatGPT, Claude, or Copilot
- But: Read it. Run it. Debug it. Explain it.

Build Understanding, not Muscle Memory

- Learning happens when you explore
- AI helps you move faster—not skip thinking



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Kick Off Questionnaire

What does AI & Computer Vision mean to you?



[https://
forms.microsoft.com/e/
KX5jKFSbPK](https://forms.microsoft.com/e/KX5jKFSbPK)

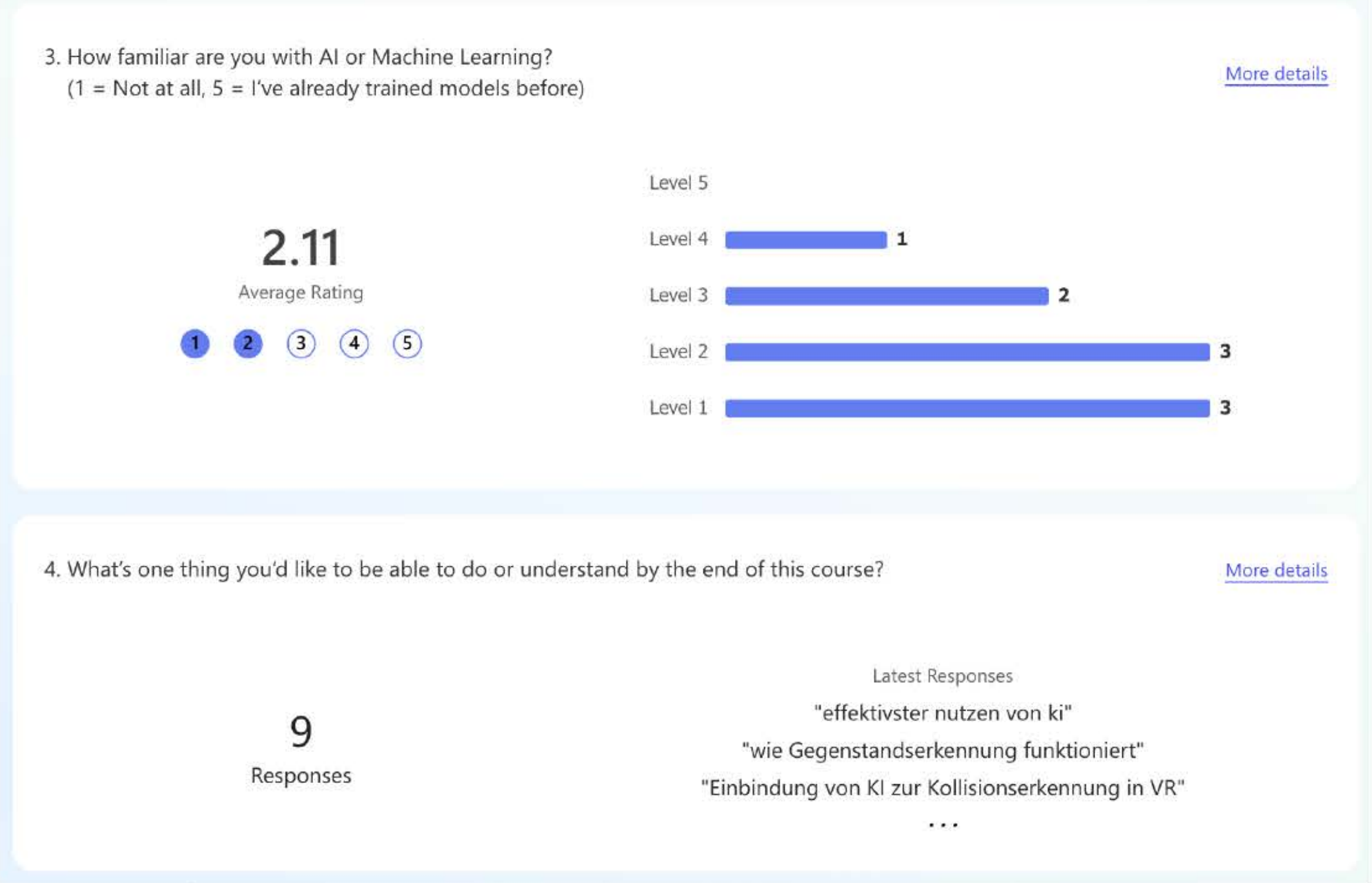
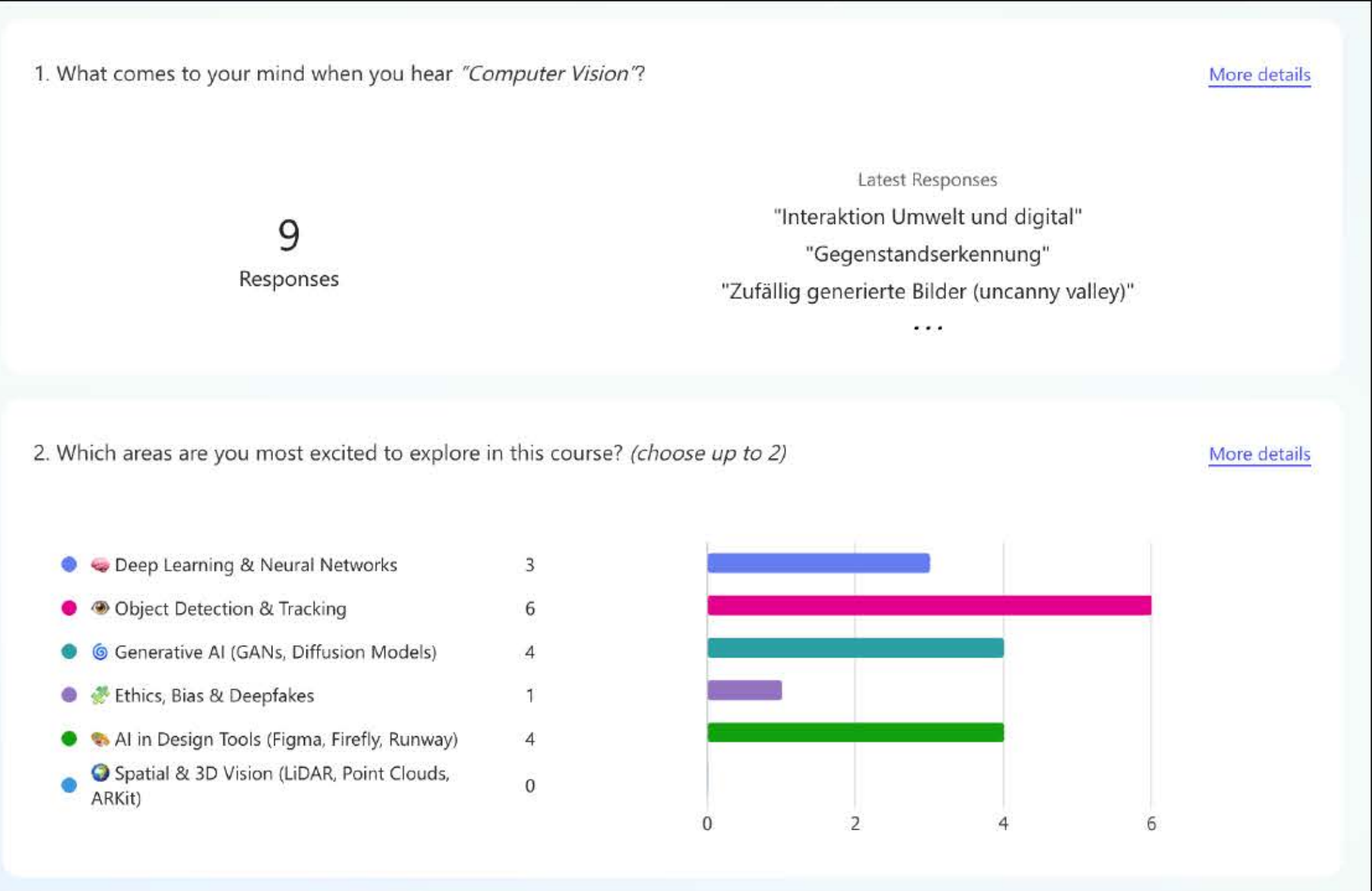
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Kick Off Questionnaire

What does AI & Computer Vision mean to you?





Introduction to Pyhsical AI: The foundation to intelligent systems

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Machine Perception (Physical AI) technologies are built to mimic human senses

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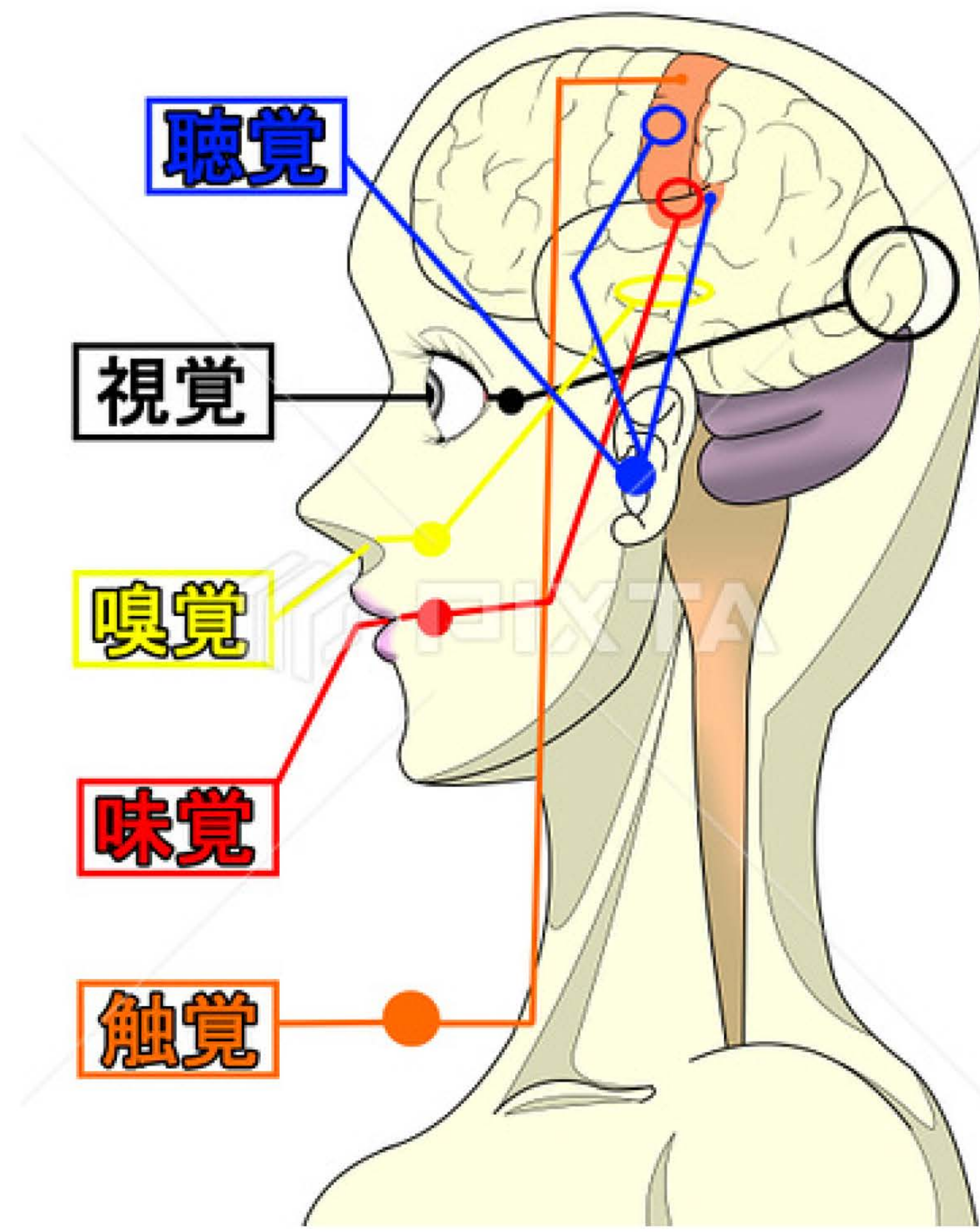


Machines that See, Hear, Feel, Smell, and Taste

- 1. Visual
- 2. Auditory
- 3. Gustatory
- 4. Olfactory

- 5. Tactile
- 6. Vestibular
- 7. Proprioception
- 8. Interoception

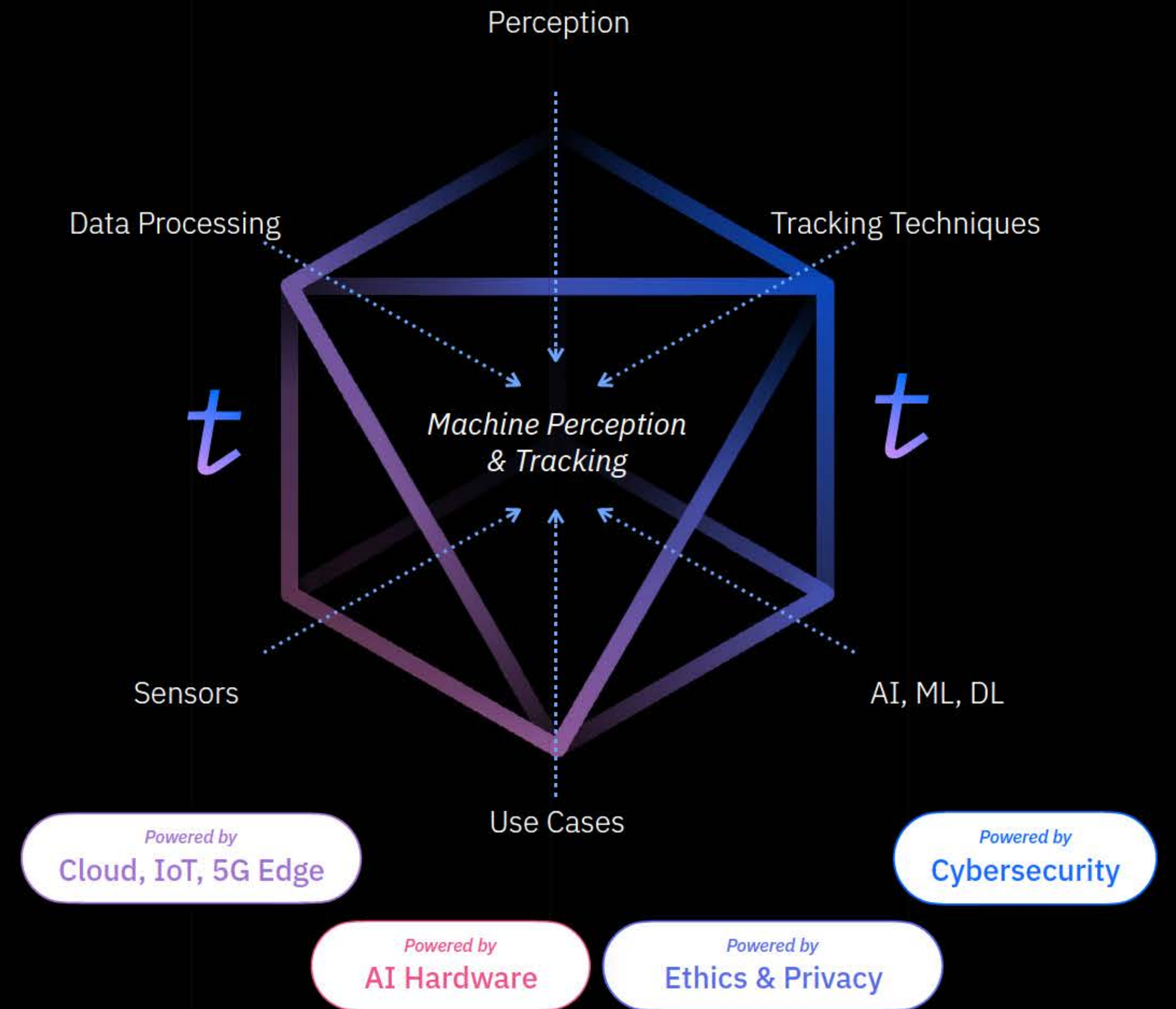
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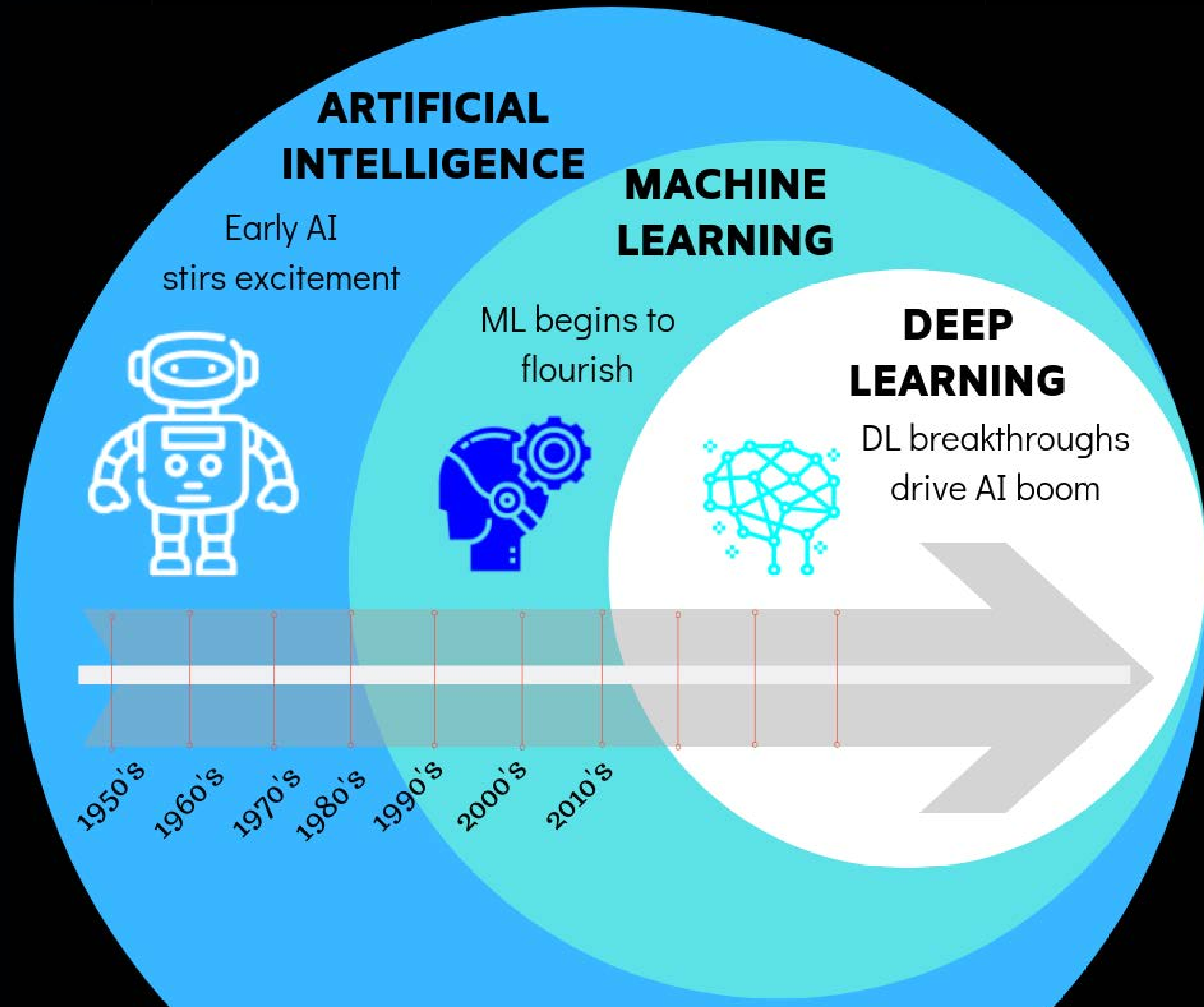
Machine Perception and Tracking is the ability of AI systems to process and interpret sensory data over time.

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Techniques



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Sensors in an iPhonePro

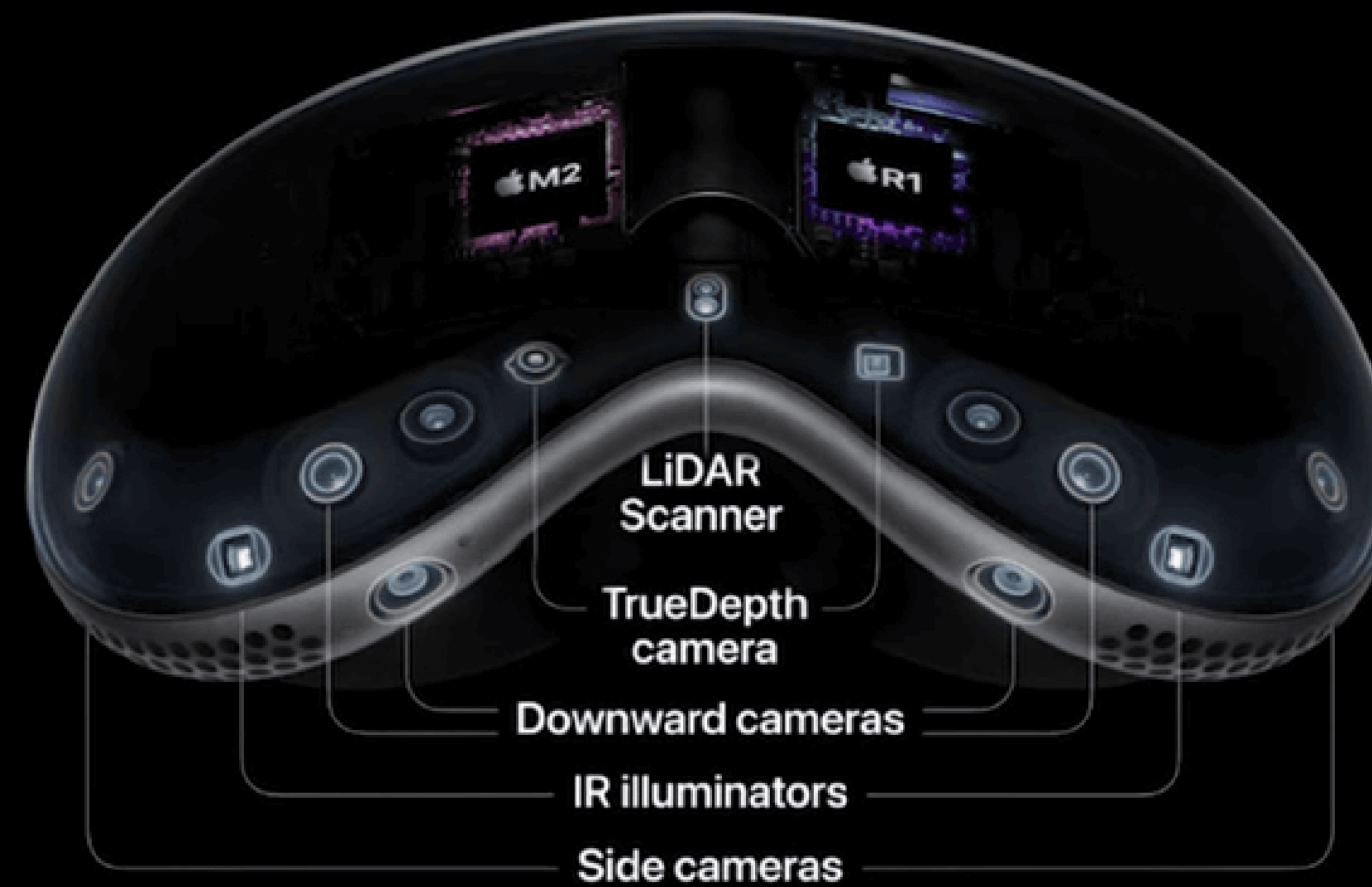
1. Cameras
2. Microphones and Loudspeakers
3. Accelerometer and Gyroscope
4. Proximity Sensor
5. Ambient Light Sensor
6. Barometer
7. Magnetometer
8. LiDAR Scanner





Sensors in Apple Vision Pro

Cameras x12.
Microphones x6
LiDAR Scanner
IR Sensors
TrueDepth Camera



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Impact and Influence of Physical AI

Tesla Robot

- allows machines to understand and interact with the world,
- automating and improving many processes.
- foundational to many technological advancements

The Eyes and Ears of AI:

- provides sensory input an AI system needs to understand and interact with its environment.
- This can greatly enhance the system's ability to operate independently and make intelligent decisions.



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Boston Dynamics



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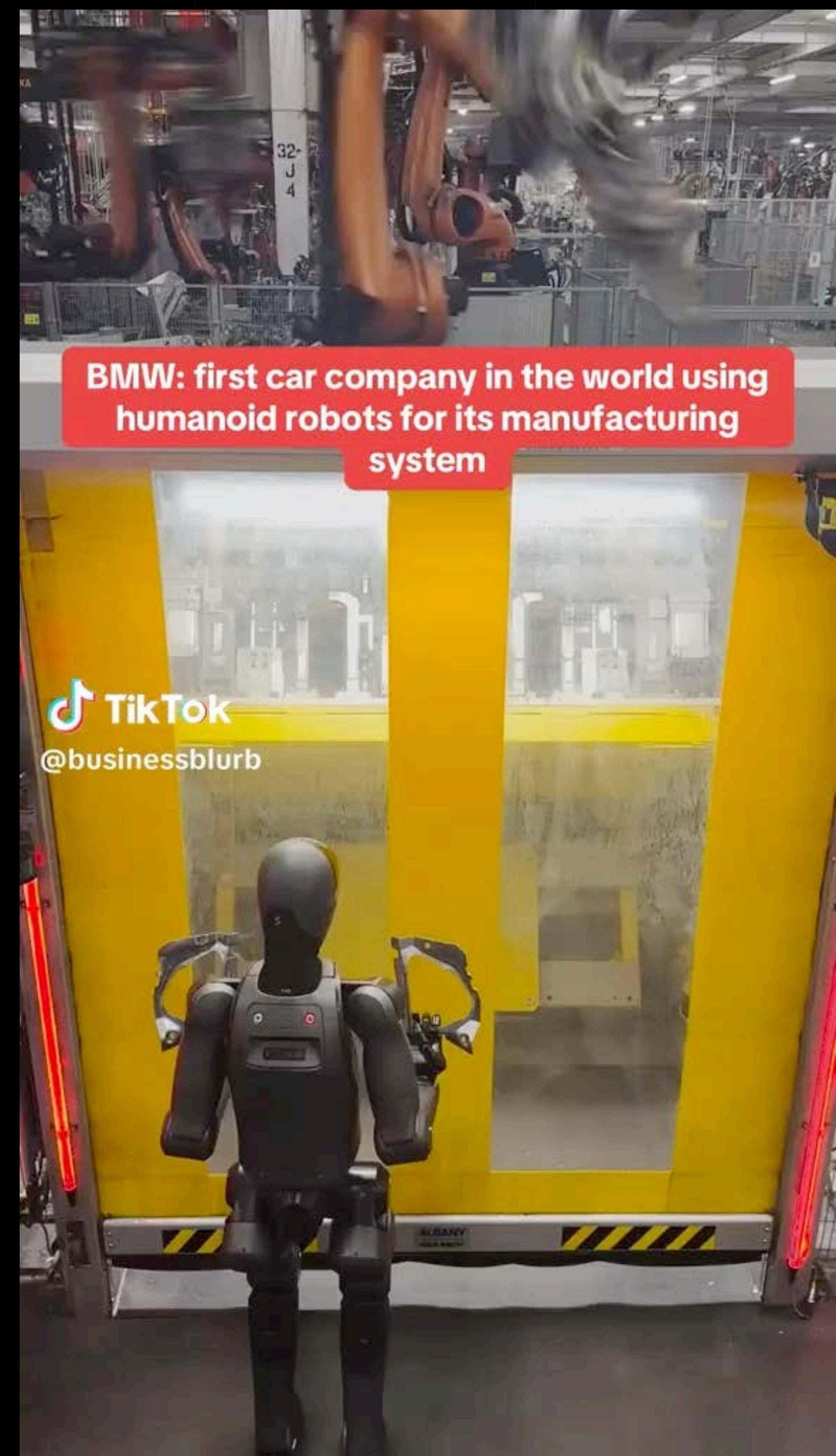


Boston Dynamics



BOSTON DYNAMICS


TikTok
@cnetdotcom



**BMW: first car company in the world using
humanoid robots for its manufacturing
system**


TikTok
@businessblurb

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IBMHypertaste



Scent



Audio/Vibration

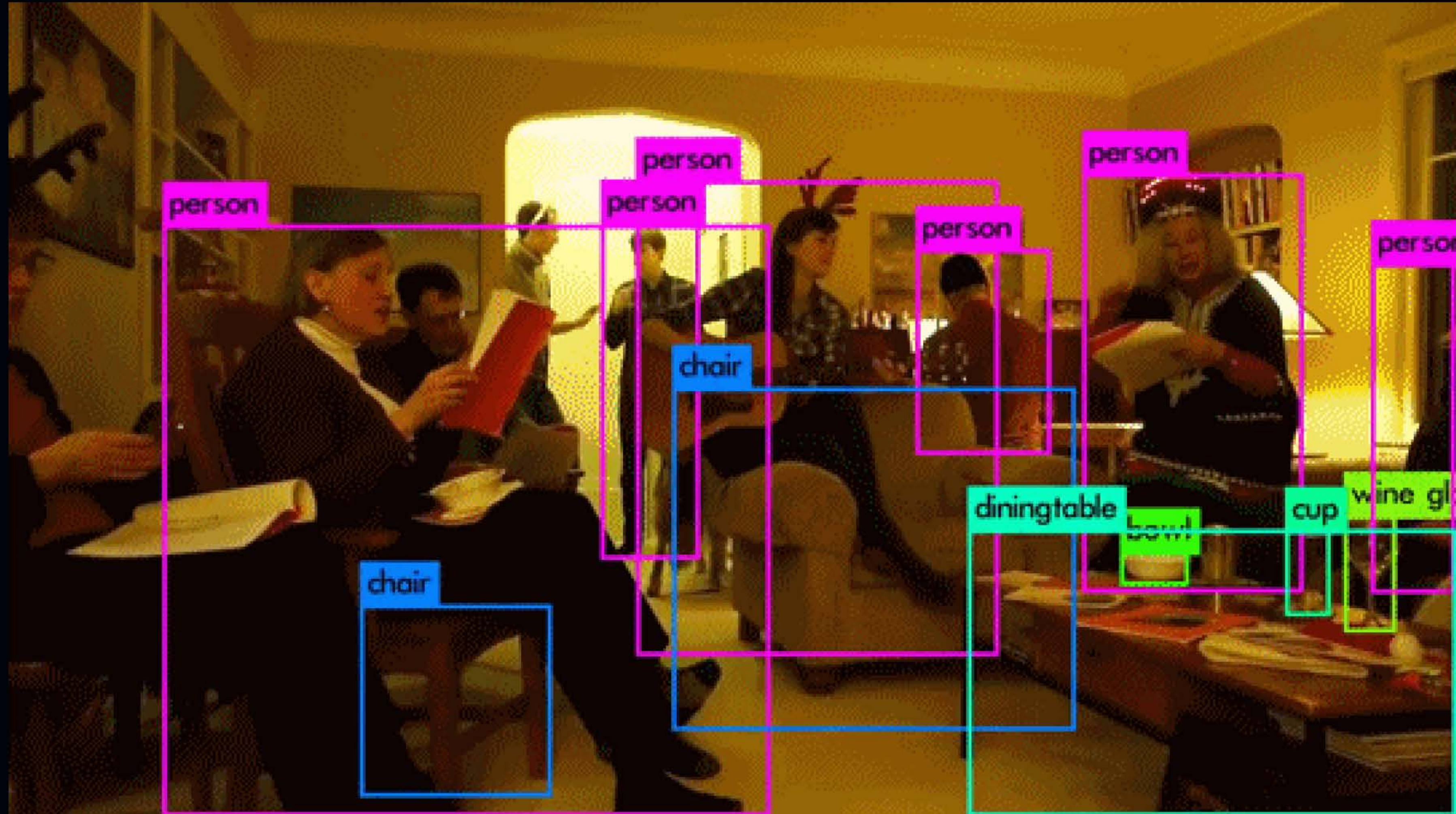


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Object Detection



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Recyclable

Non-Recyclable

Recyclable

Non-Recyclable



TheBarcode Scannerasan examplefor oneofthefirst CVProject

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Techniques

ImageProcessing

filtering, edgedetection, coloranalysis, andnoise reductiontopreparethe inputdataforfurther analysis.

FeatureExtraction

identifyandextractsignificant patternsorfeaturesfromraw data.identifyingedges, corners,andtexturesin imagesorrecoznizingspecific frequenciesinanaudio

DepthEstimation

Determiningthedistance fromthesensortoeach pointinthescene.

MotionEstimation

Calculatingthemovementof objectsintheenvironment.

MachineLearning

Supervisedand andreinforcementlearningto enableamodeltolearnfrom dataandmakepredictionsor decisionswithoutbeing explicitlyprogrammedto performthetask

ReinforcementLearning

combinesdeeplearningwith reinforcementlearning, enablingAISystemstolearn fromtheirenvironment throughinteractions.hasbeen usedtoteachAISystemsto playvideogames,control robots,andmorebasedon sensoryinputs.

NeuralNetworks

Mimicksthehumanbrain,CNNs learningmodelthat's particularlyeffectiveforimage andvideoprocessingtasks. Theycanlearnhierarchical featurerepresentationsindata, usefulforcomplextaskssuchas objectrecognition,scene understanding,andsemantic segmentation.

TransferLearning

usingapre-trainedmodelona newproblem.allowsusto leveragethevastamountsof pre-existingmodelstrainedon largedatasetsandadaptthem tospecifictaskswithmuch fewerdata.

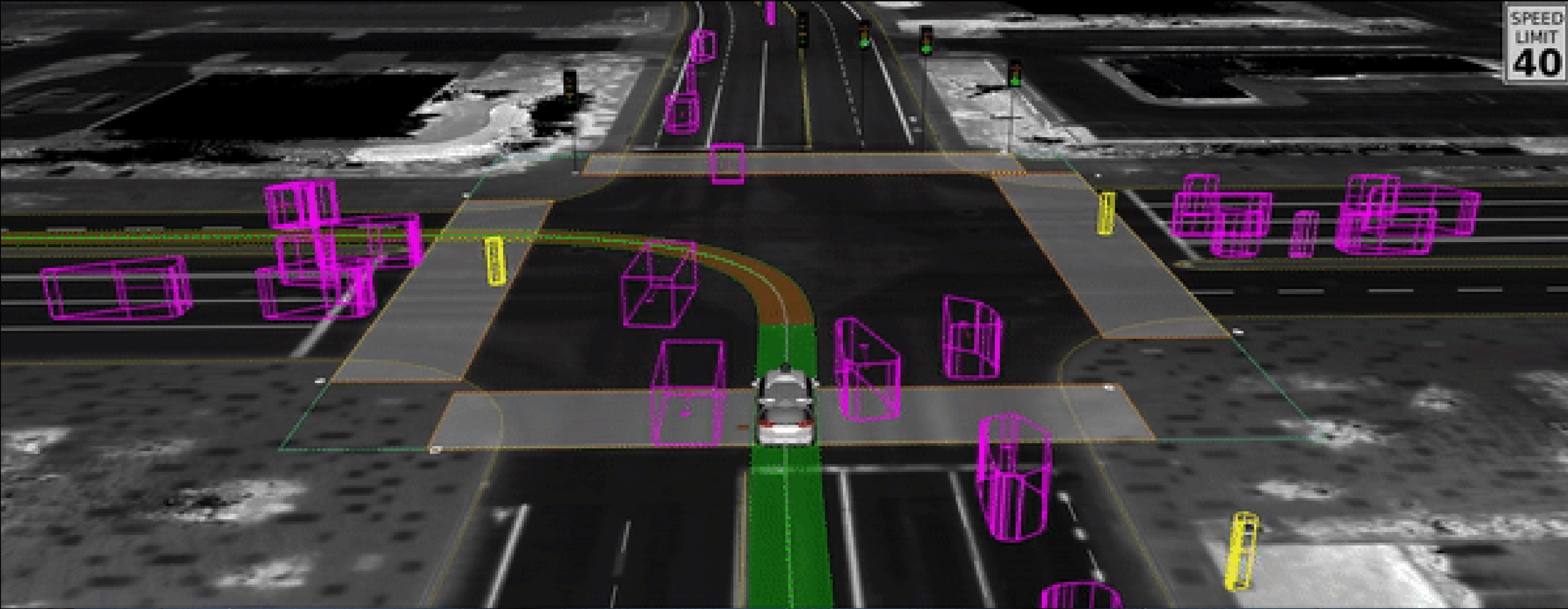
GenerativeModels

Generativemodelslike GenerativeAdversarial Networks(GANs)and VariationalAutoencoders (VAEs)generatenewdata samplesthatresemblethe trainingdata.

Thesemodelshavegenerated realisticimages,enhanced imageresolution,andmore.

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ArticulatedBodyTracking

Techniques

- PoseEstimation:Methodslike OpenPose,PoseNet,and DeepPose.
- SkeletalTracking:Constructinga skeletonfromdetectedkeypoints, andtrackingthemovementofthis skeletonacrossframes.
- DeepLearning:Techniqueslike ConvolutionalPoseMachines, StackedHourglassNetworks,and PartAffinityFields.

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Cognitive Highlights



TOTAL CLIPS PROCESSED: 7 HOURS OF COVERAGE: 14 **IBM**



CURRENT TIME: 5:28 PM MOST RECENT: 8:20 PM
MATCH POINT: Williams Williams v Shvedova Babos Doubles Final Highlights



8:20 PM
MATCH POINT
0.76
EXCITEMENT LEVEL
[SEE FULL HIGHLIGHT VIDEO](#)



3:15 PM
SHAPOVALOV WITH THE
BREAK
0.23
EXCITEMENT LEVEL
[SEE FULL HIGHLIGHT VIDEO](#)



1:41 PM
FIRST SET
BOLKVADZE/MCNALLY
WITH THE BREAK
0.25
EXCITEMENT LEVEL
[SEE FULL HIGHLIGHT VIDEO](#)

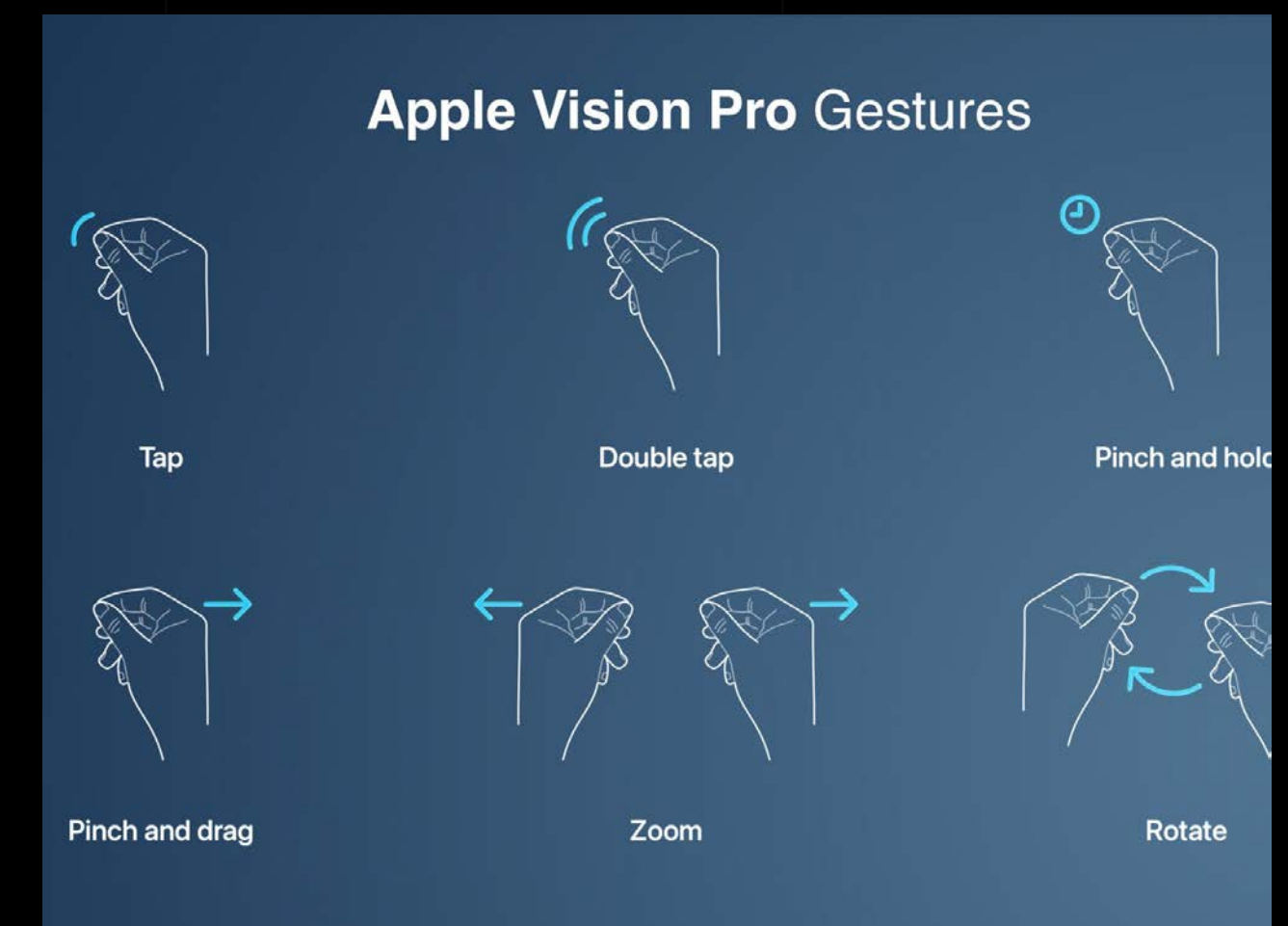
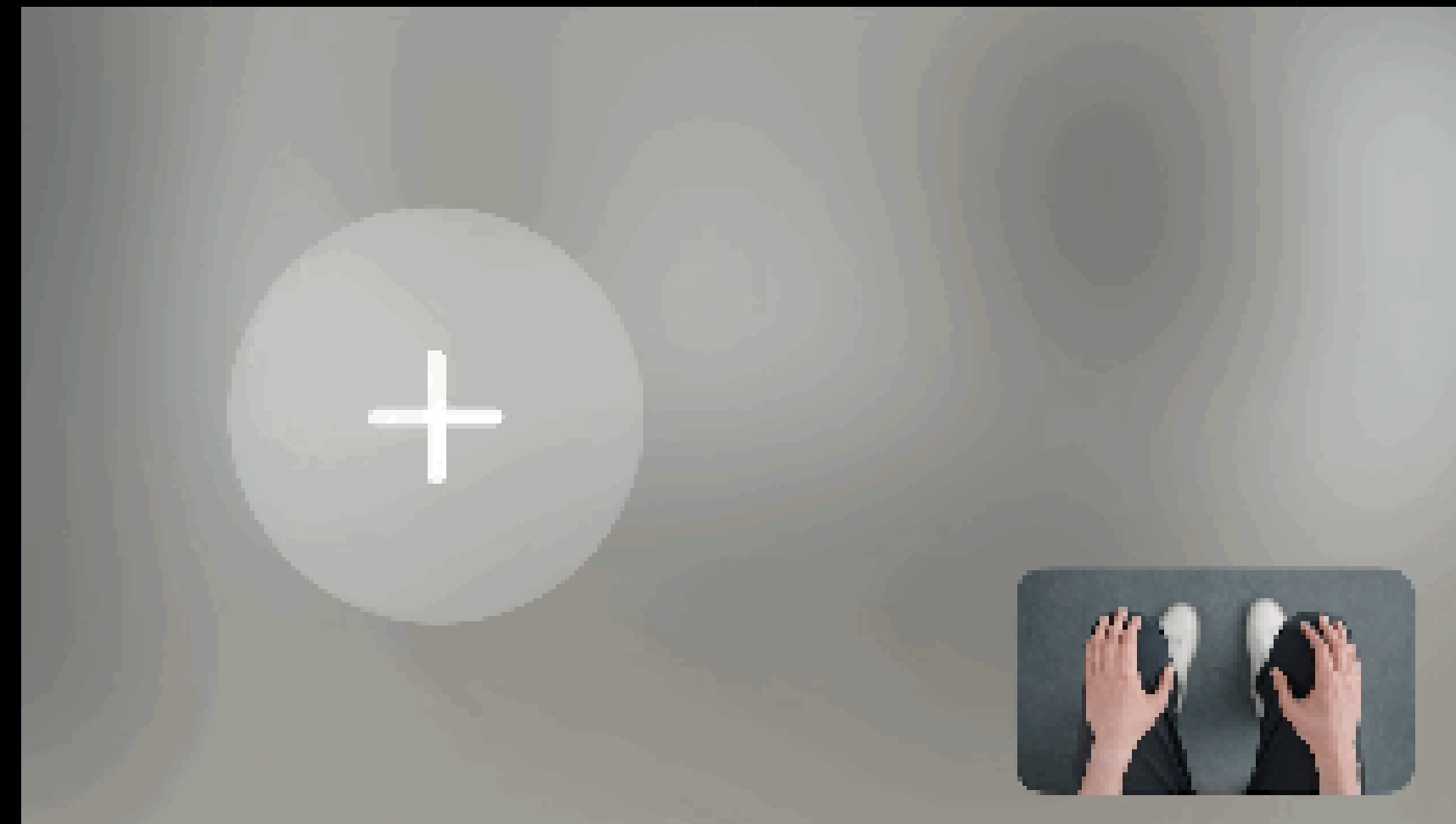


2:40 PM
MATCH POINT
0.51
EXCITEMENT LEVEL
[SEE FULL HIGHLIGHT VIDEO](#)



GestureRecognition

[https://
developer.apple.com/
documentation/vision/
detecting_hand_poses_
with_vision](https://developer.apple.com/documentation/vision/detecting_hand_poses_with_vision)



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< 3:24 Done

Hand Gestures

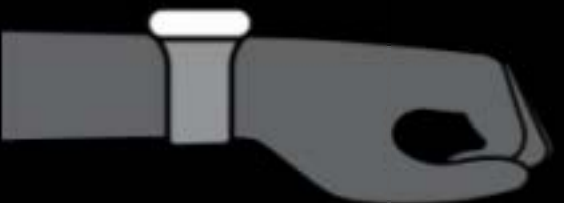
- Pinch
- Double Pinch
- Clench

3:25 Done



Pinch
Try tapping your pointer finger and thumb.

3:25 Done



Clench
Try closing your whole hand to form a fist.

3:25

✓

Clench Recognized
Close your hand into a fist one more time to return to Hand Gestures.

REAL FOOTAGE OF MUDRA BAND IN ACTION



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Gesture Recognition inPractice

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the sign language interpreter
was not fking around 🤔





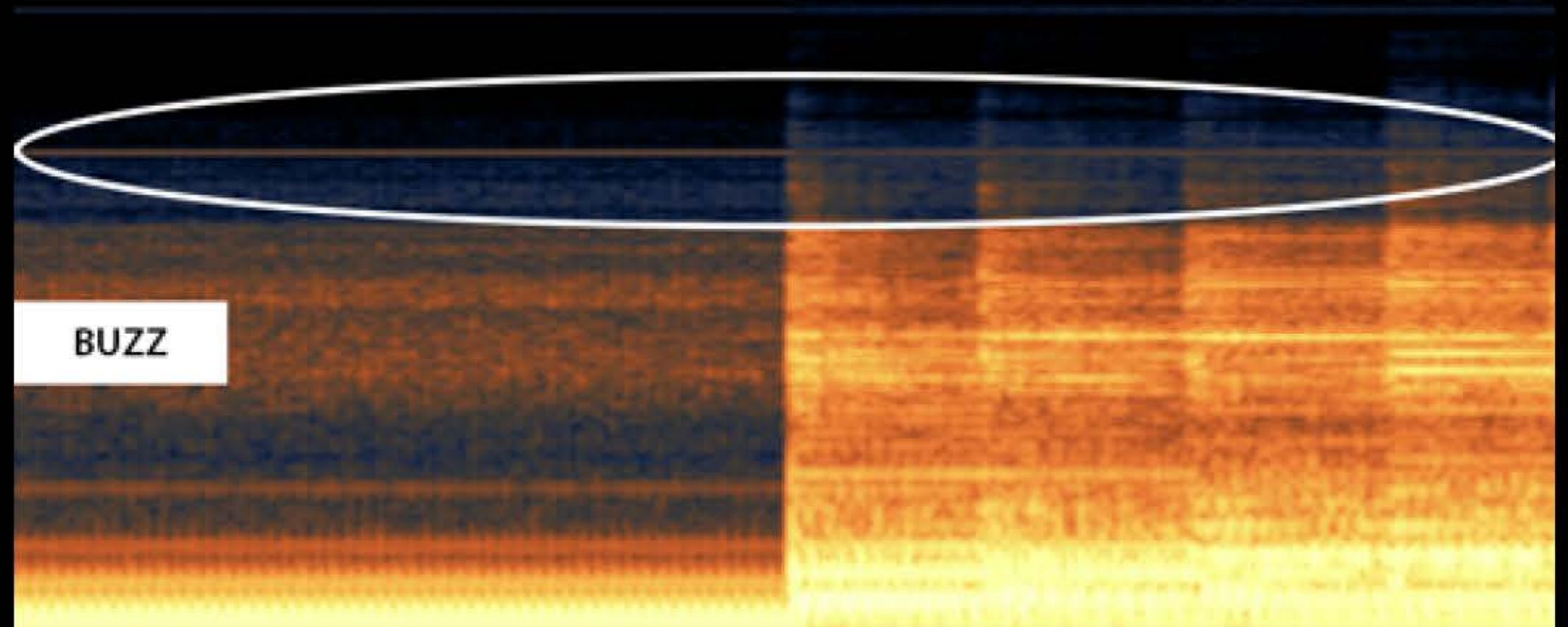
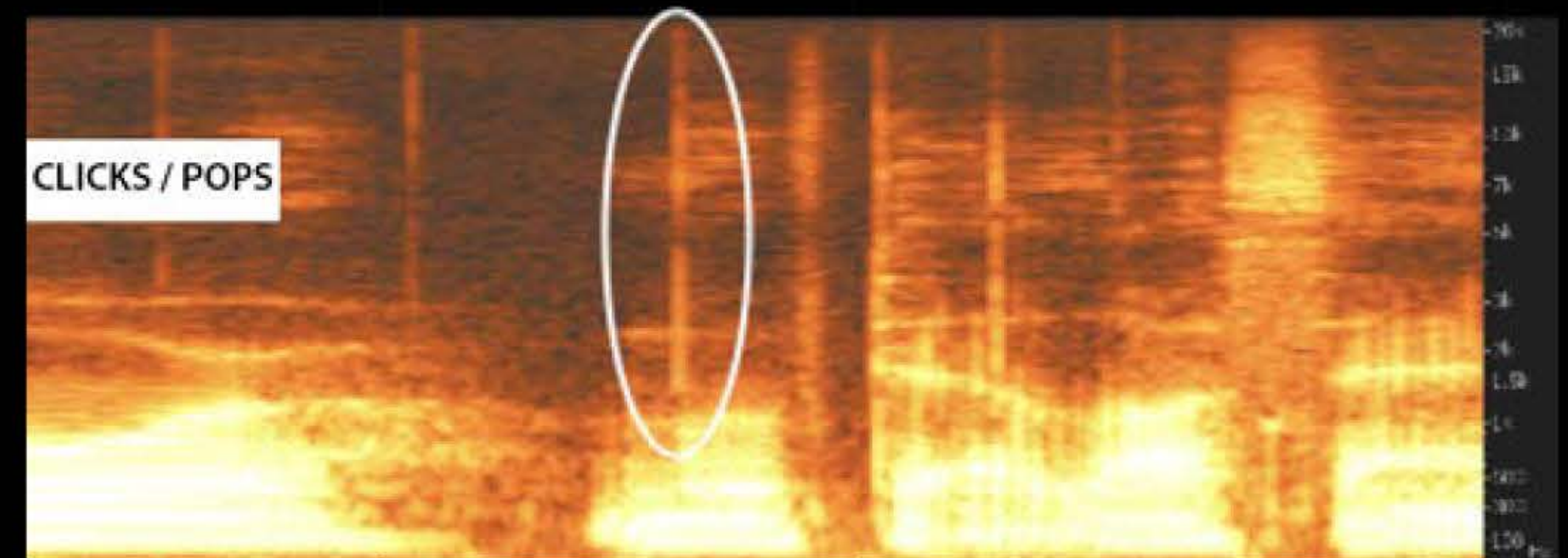
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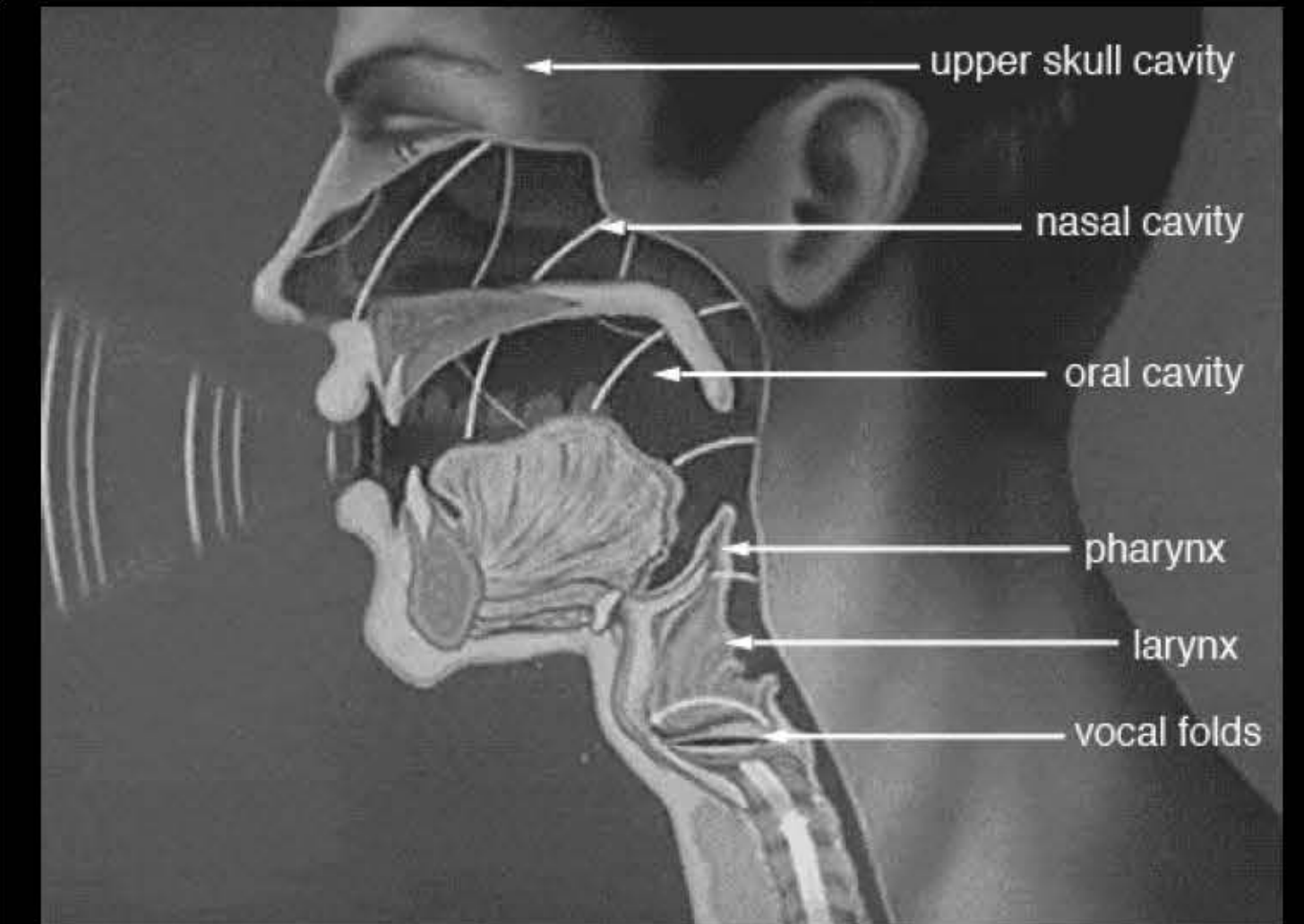
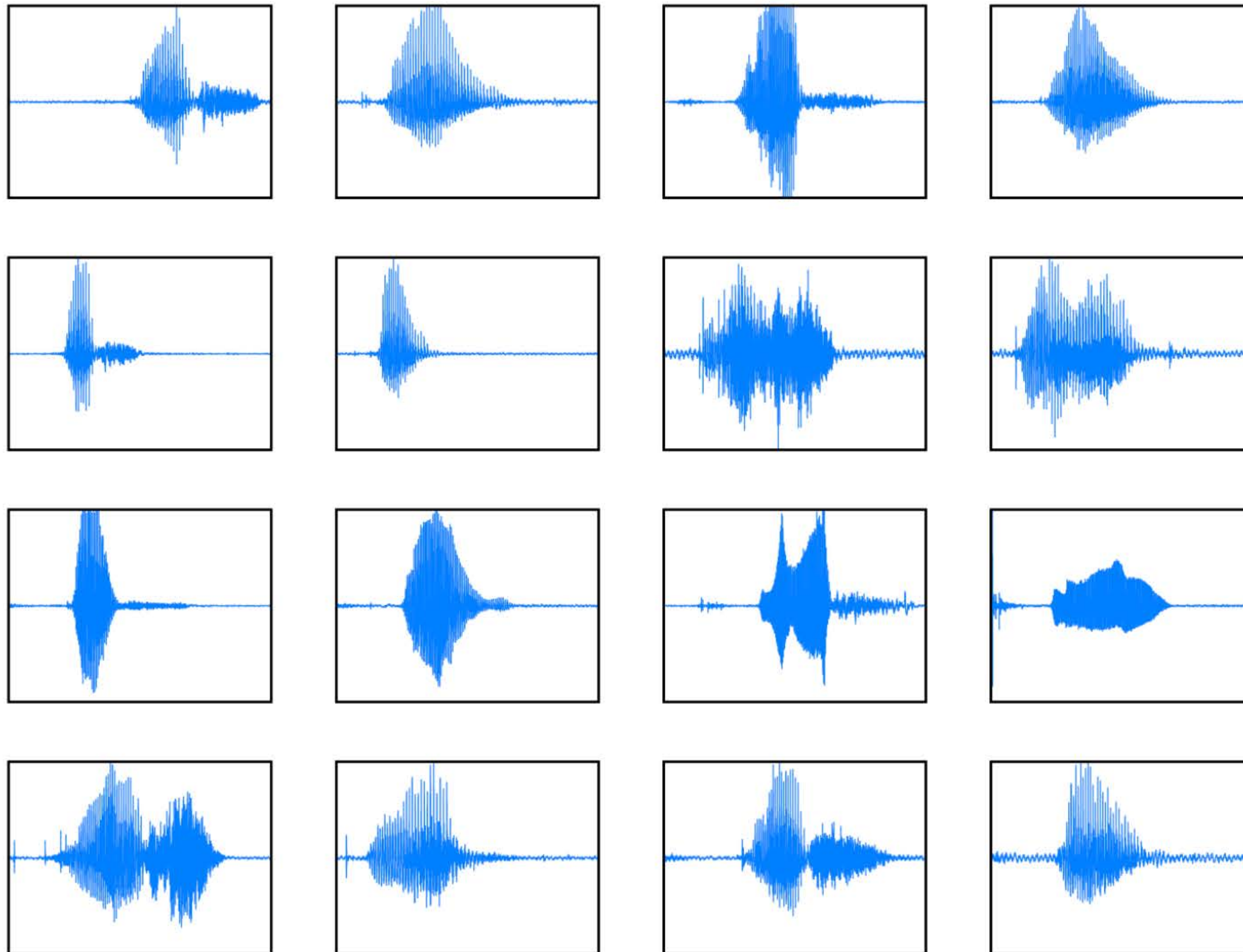
Using the Spectrogram to solve Audio problems

- Hum
- Buzz
- Hiss and other broadband noise
- Clicks, pops, and other short impulse noises
- Clipping or distortion
- Intermittent noises
- Gaps and drop outs





A Thousand Times “No”

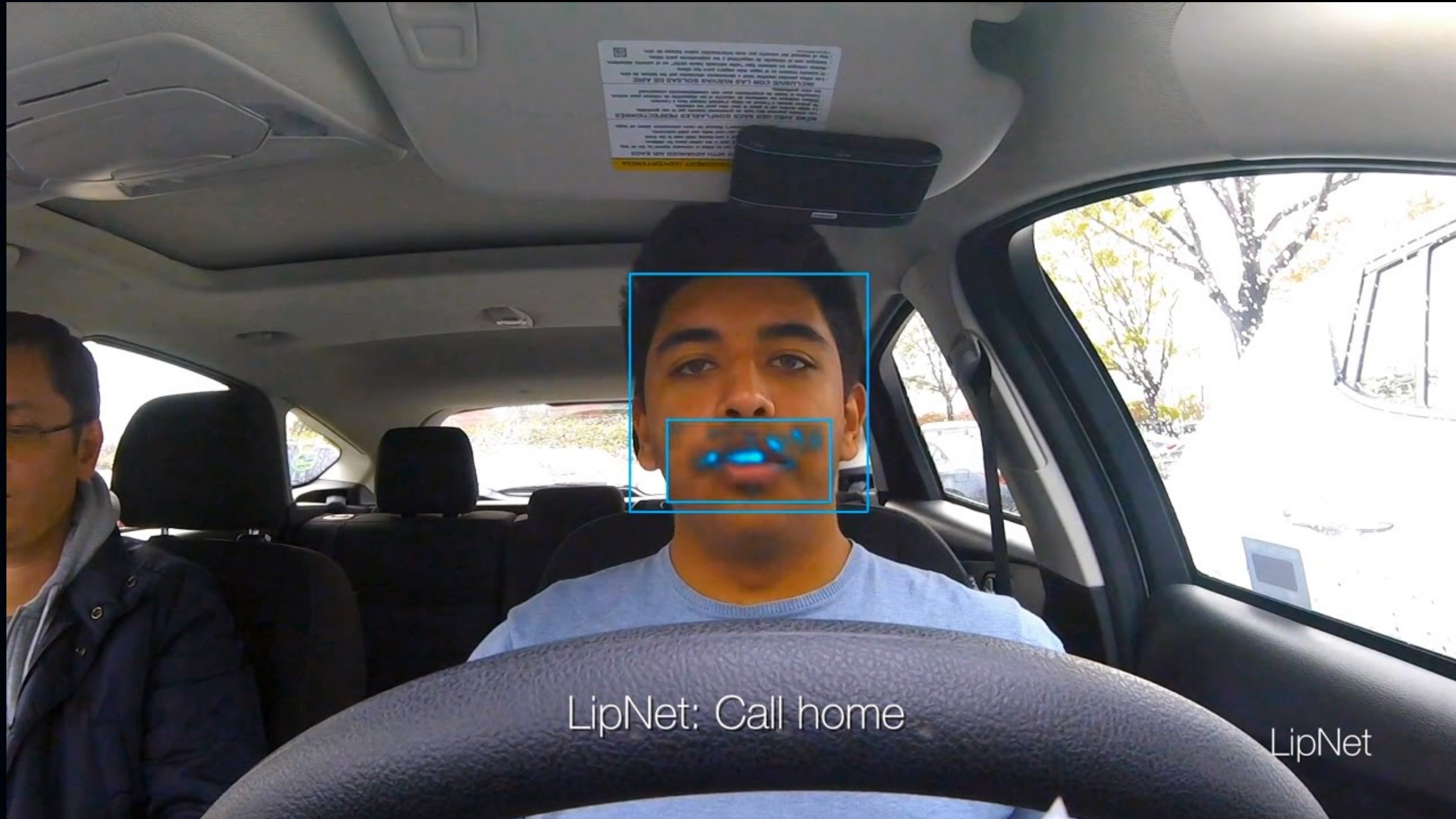


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Combining Visual & Audio data for Audio-Visual-Speech-Recognition (AVSR)



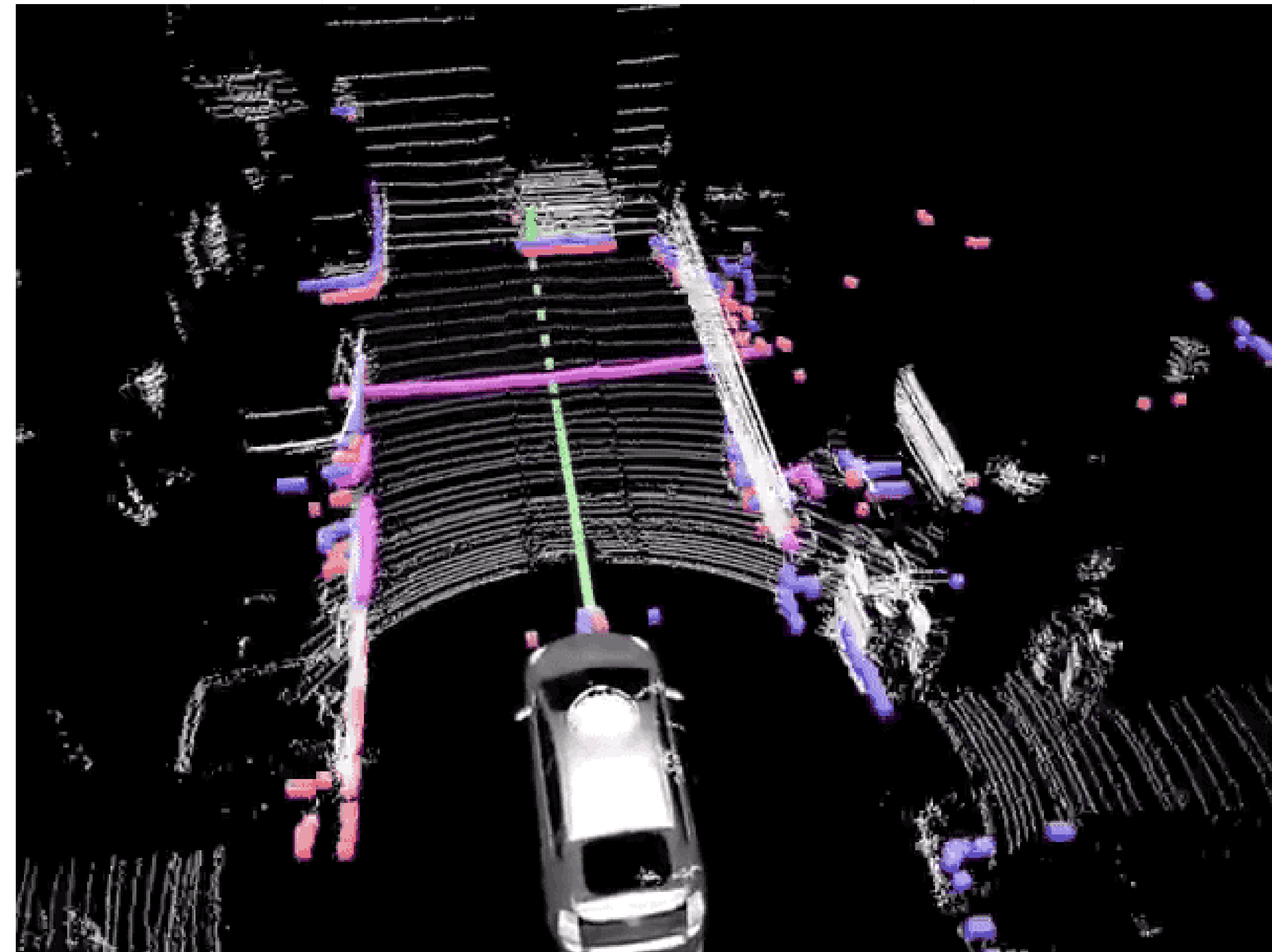
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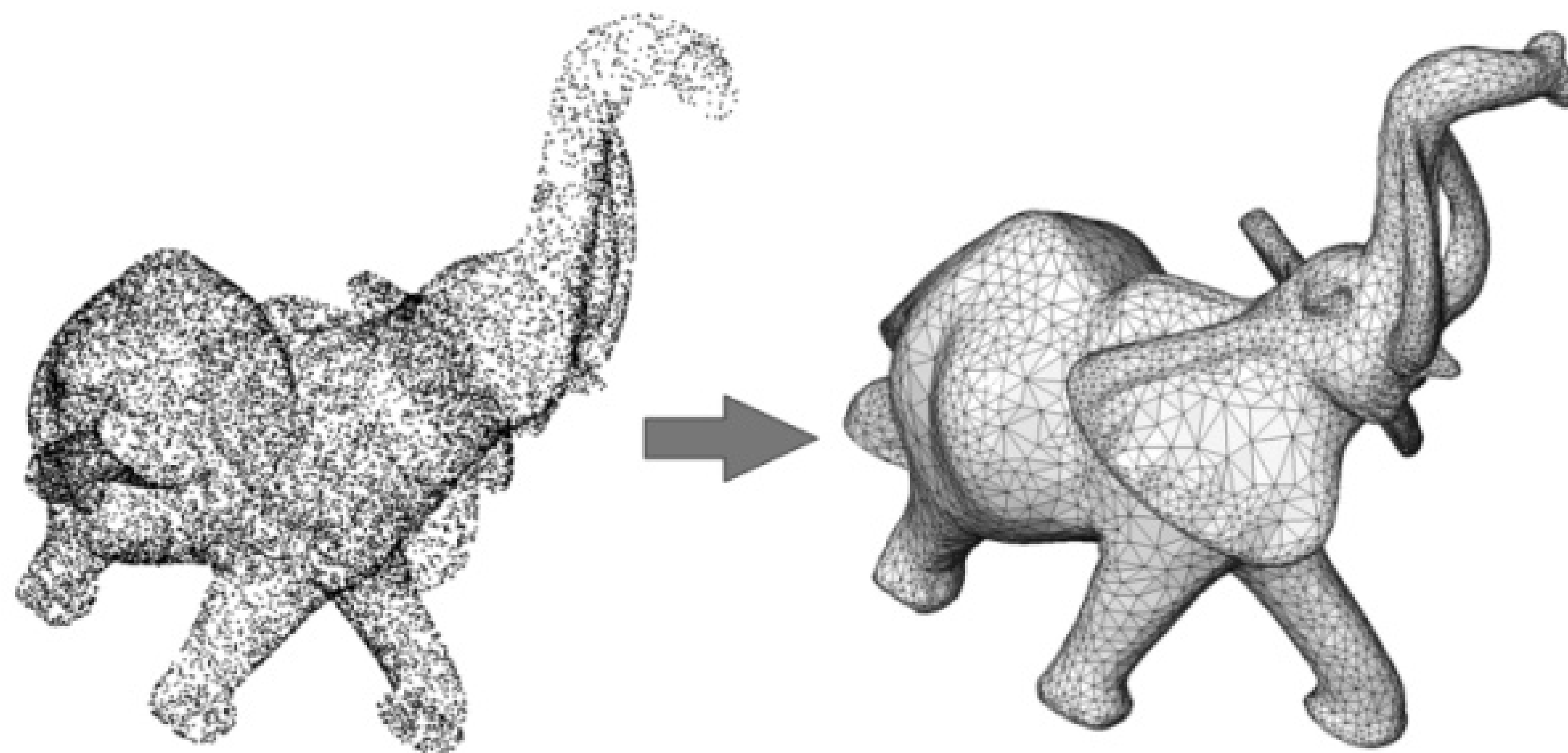
LIDAR is a technology built into self-driving cars (or your iPhone and iPad). In cars it ranges from 100m-300m (depending on the sensor)

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LiDARcapturedPointCloudSurface Reconstructionto create3Dmodels





PointCloud Segmentation

- Pointcloudsegmentation:Partitioningapointcloud
intomeaningfulsegmentsbasedonsemanticor
geometricproperties

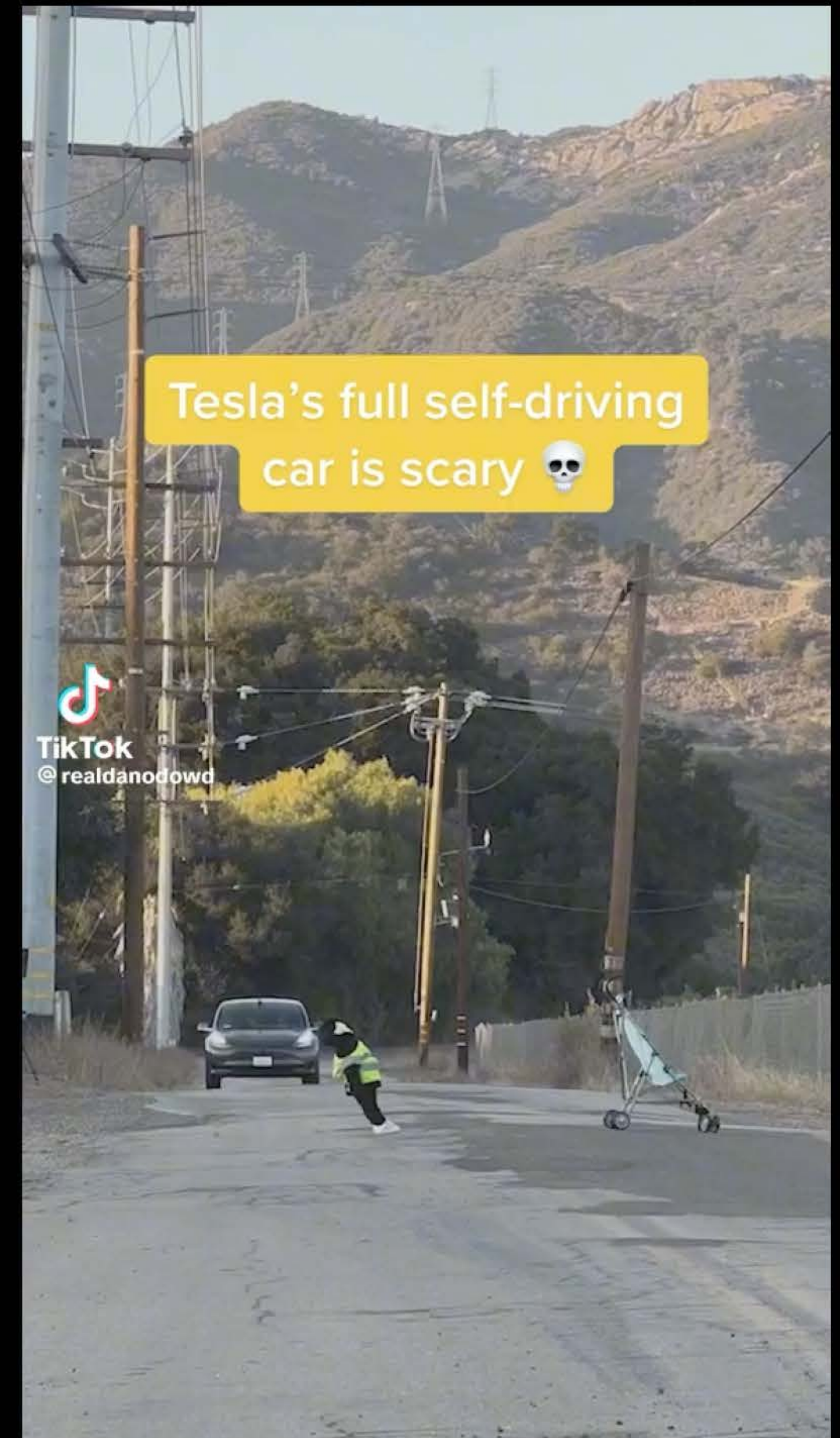
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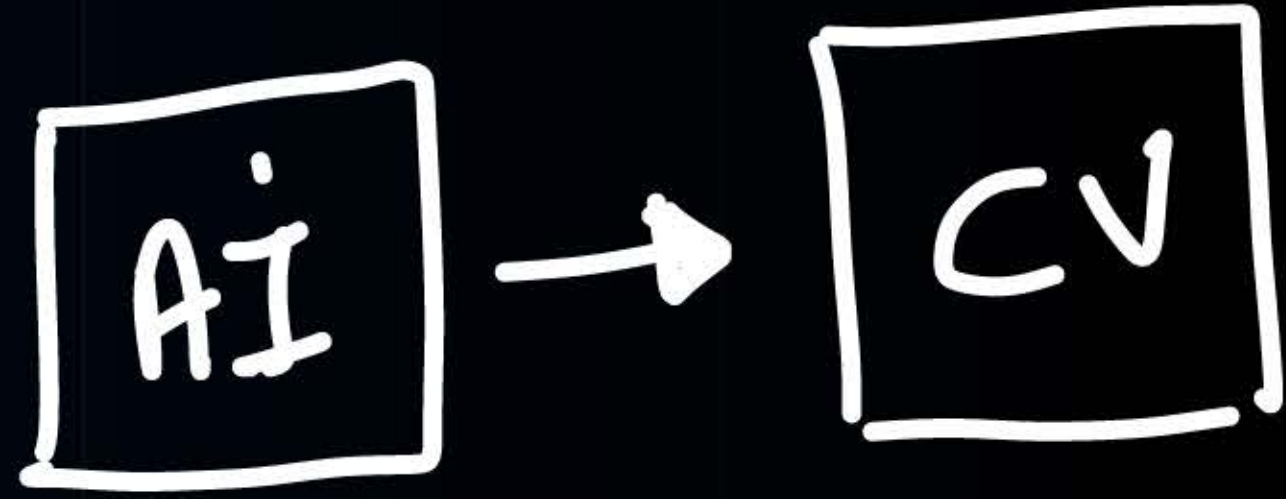




These days we expect things just to work, and without data fusion, it wouldn't.

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Computer Vision (CV) **Trains Computers to Interpret & understand The Visual World.**

Computer Vision (CV) can be your way in to AI as Designers & Front-End Developers – Just try it out!





Python Libraries & their usage

Python

A high-level, interpreted programming language known for its readability and vast range of libraries

OpenCV

An open-source library aimed at real-time computer vision. It includes several hundreds of computer vision algorithms.

Matplotlib

A plotting library for Python, useful for creating static, animated, and interactive visualizations in Python

NumPy

A library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions

TensorFlow/ Keras

Open-source libraries for machine learning and neural networks, TensorFlow and its high-level API Keras will be used for creating models for object detection and image/sound classification

Open3D

Open-source library to process LiDAR data, Point Clouds and create Surface Reconstruction and other 3D image manipulation

Mediapipe

Gesture & Face Recognition with landmarks by Google

YOLOvX

Real-time object detection

...

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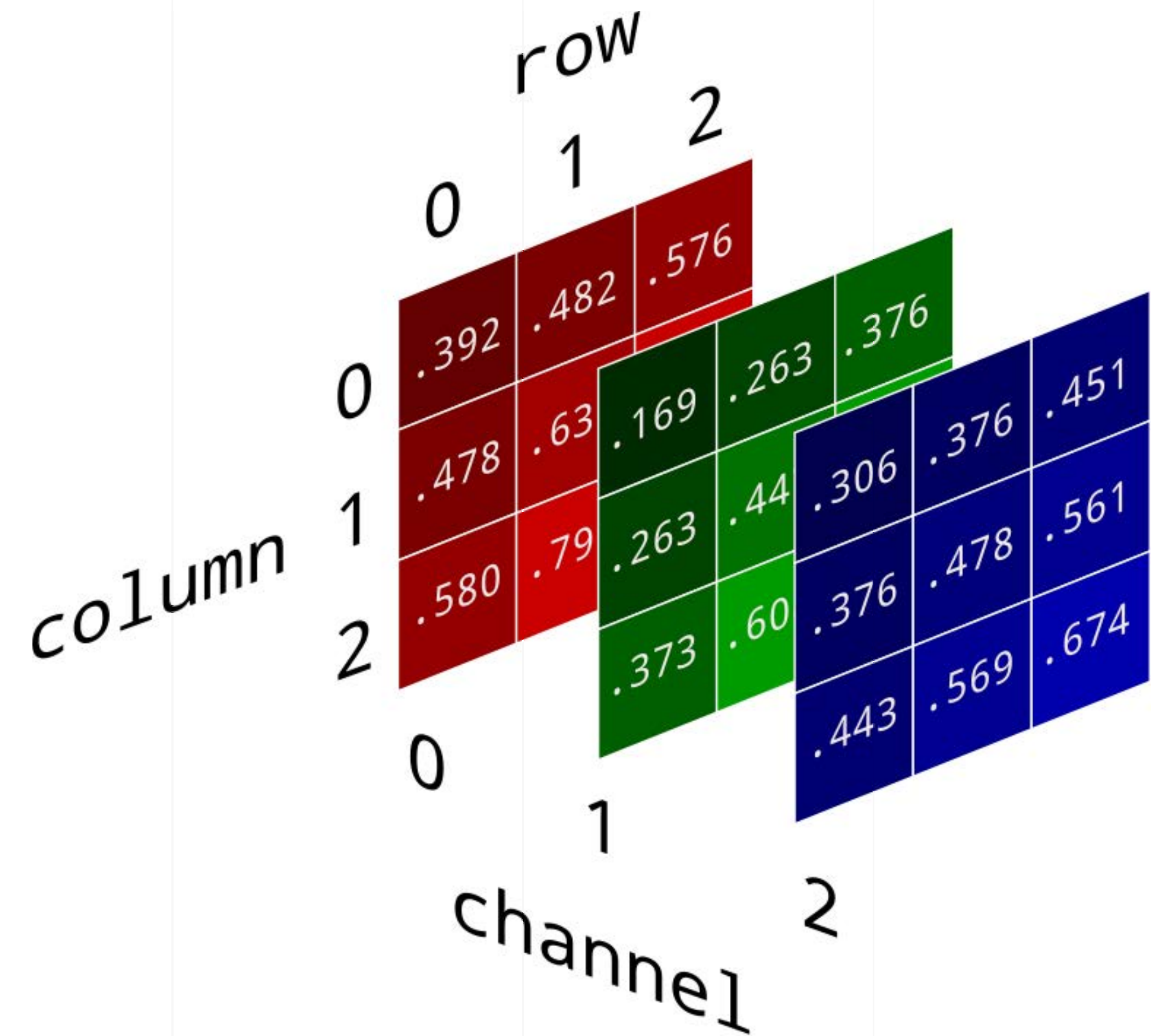




Animageisan
arrayormatrix,
consistingof3
channelsandrows
+columns

For greatest flexibility we can pass Image to numpy
array to do scientific operations on it.

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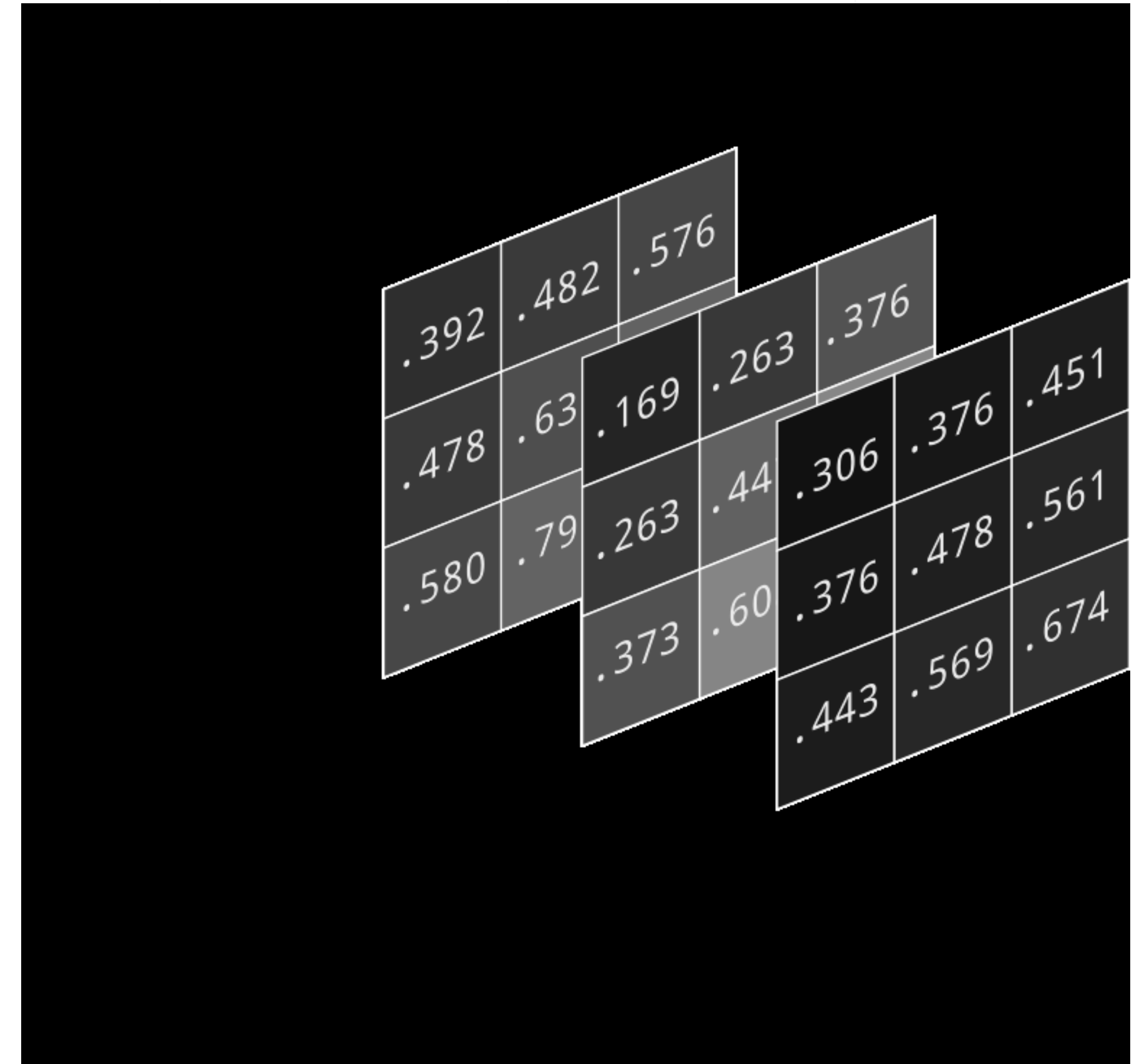




Machines are Color-Blind to process images faster.

First step in all algorithms is to make an image
greyscale so that the algorithm only deals
with 1 channel instead of 3

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Data Augmentation techniques allow models to adapt and work better in a real environment.

Original dataset 9,000 images

Dataset increased 40,000 images

Saturation
Contrast
Rotate
Sharpness
Fuzziness
Tonality
Noise
Cut
Normalize
Contrast
Brightness

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IBM Confidential

Example DataGenerator with Keras

```
from keras.preprocessing.image import ImageDataGenerator

image_gen = ImageDataGenerator(rotation_range=30, # rotate the image 30 degrees
                                width_shift_range=0.1, # Shift the pic width by a max of 10%
                                height_shift_range=0.1, # Shift the pic height by a max of 10%
                                rescale=1/255, # Rescale the image by normalizing it.
                                shear_range=0.2, # Shear means cutting away part of the image (max 20%)
                                zoom_range=0.2, # Zoom in by 20% max
                                horizontal_flip=True, # Allow horizontal flipping
                                fill_mode='nearest' # Fill in missing pixels with the nearest filled value
                                )
```

Example Data Augmentation Effects



Original image without data augmentation



Magnified image with low blur



Magnified image with high level of blur

1

Results with Original Data Set



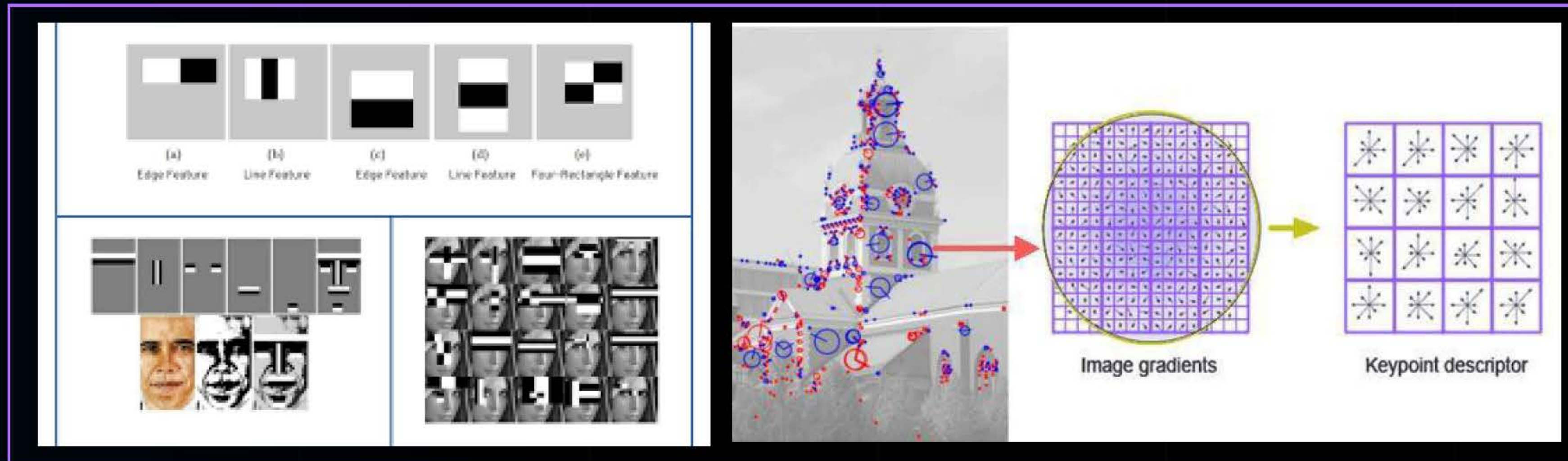
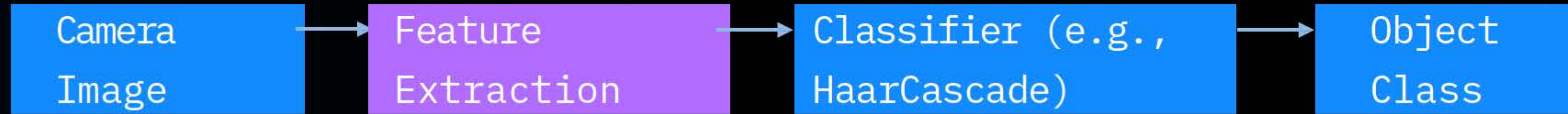
Results with Augmented Data Set



horby



Computer Vision Pipeline – Classical CV / Data-Driven Approach





Computer Vision Tasks

Classification



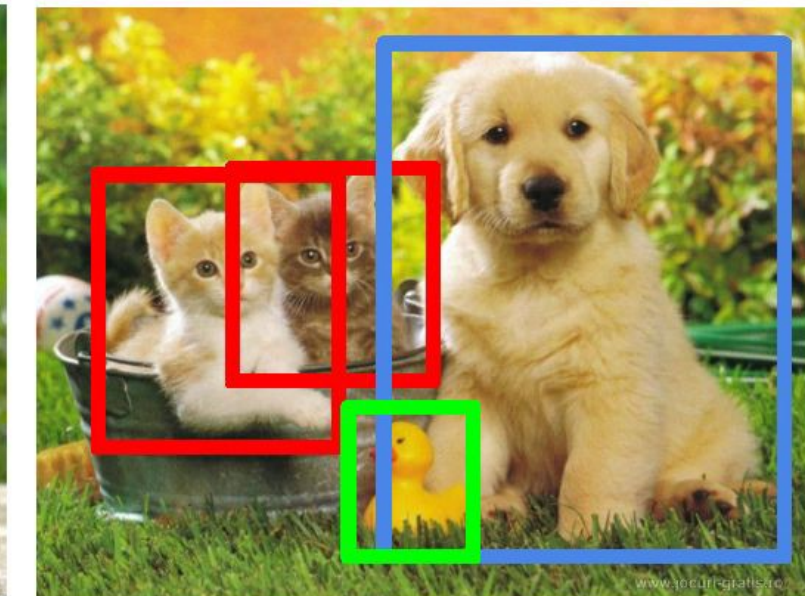
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

Instance Segmentation



CAT, DOG, DUCK

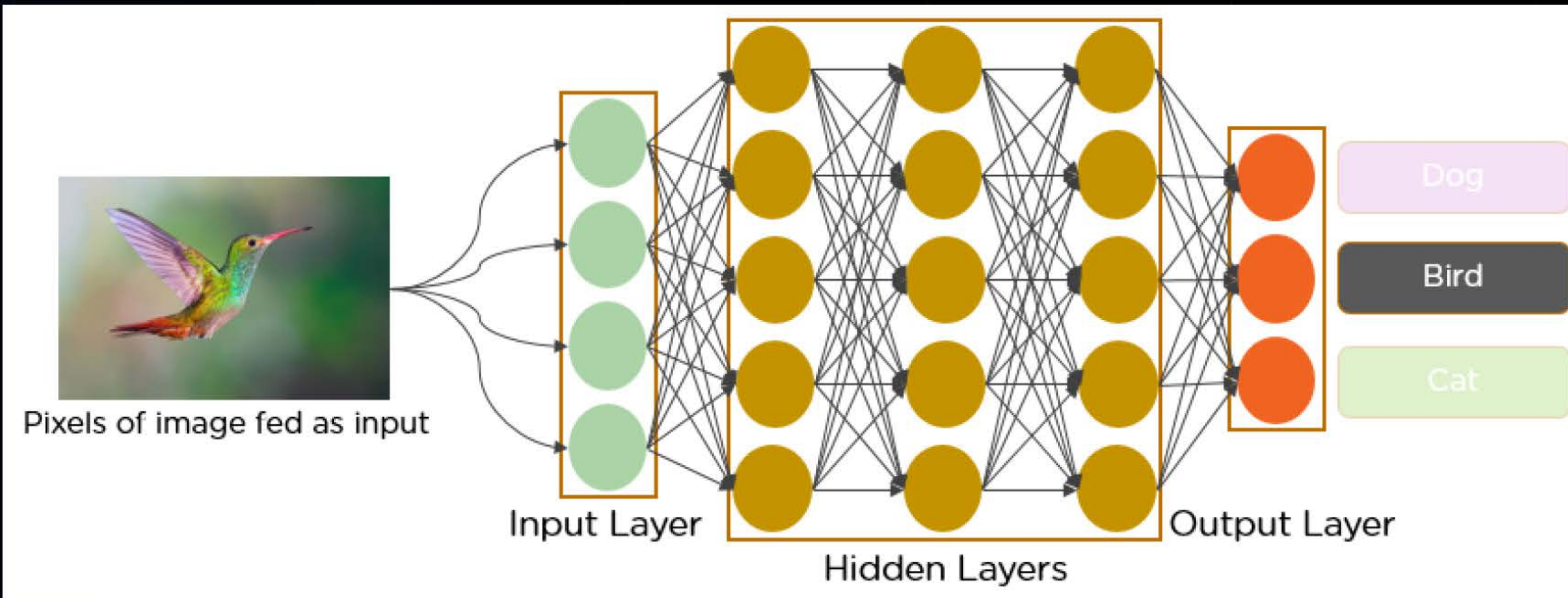
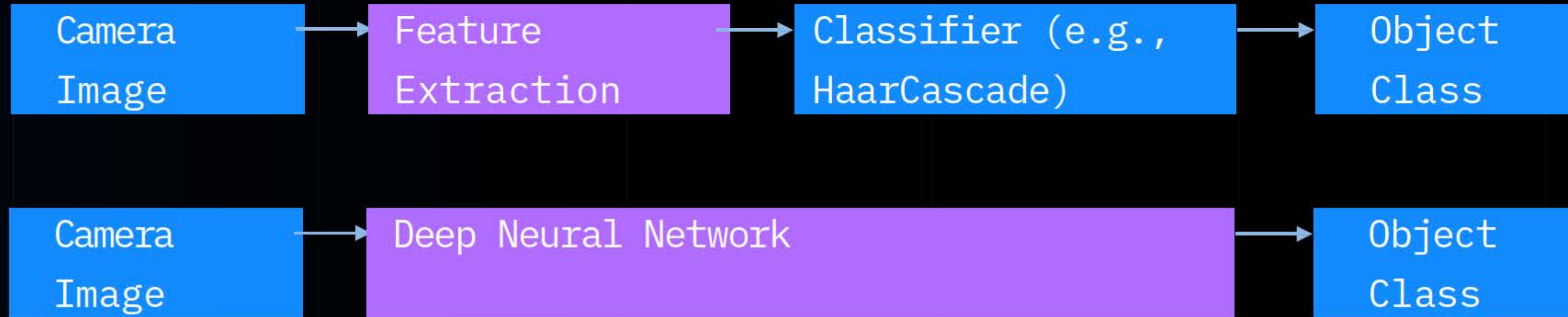
Single object

Multiple objects

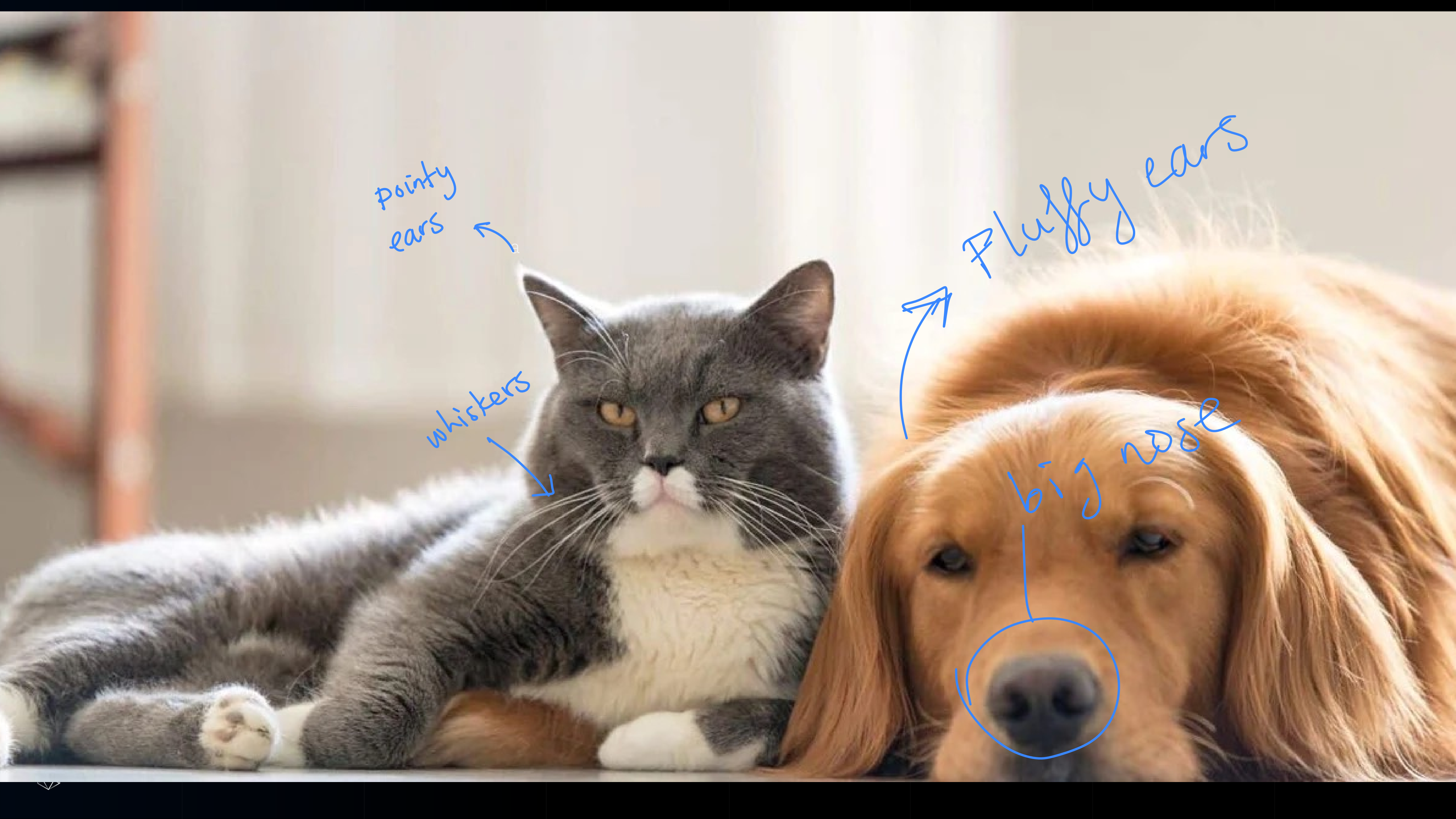




Deep Learning Computer Vision Pipeline



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pointy
ears

whiskers

fluffy ears

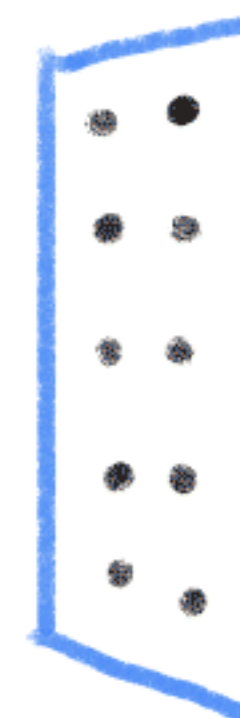
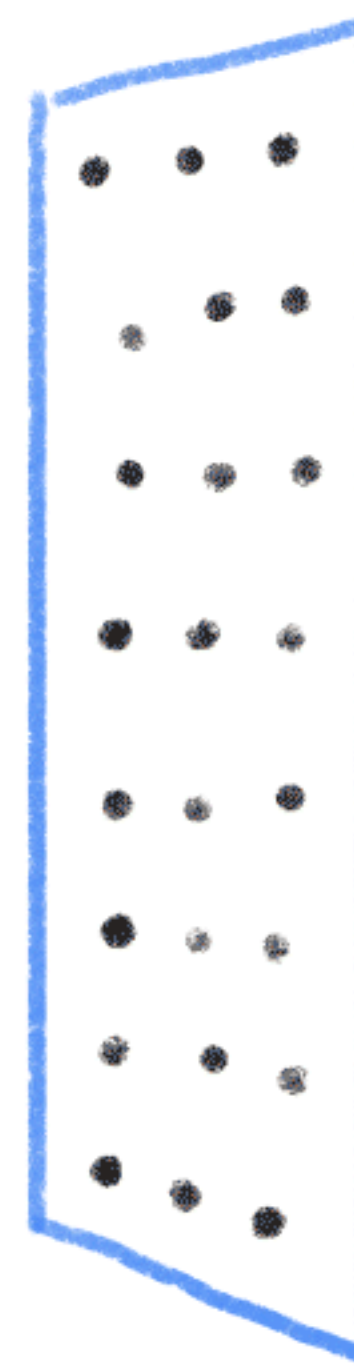
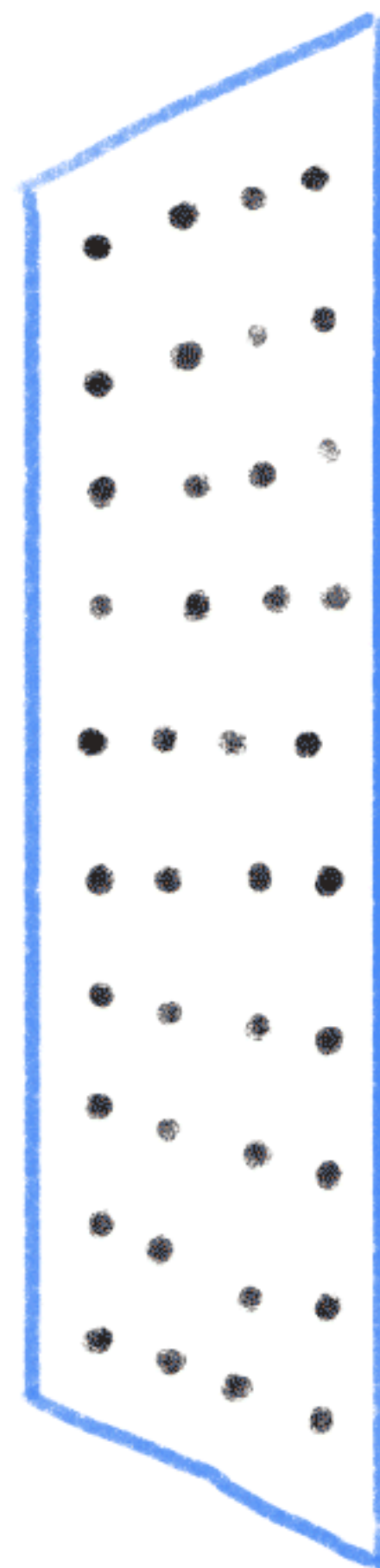
big nose

CAT



(LBELED
PHOTOS)

DOG



OUTPUT



TeachableMachine

<https://teachablemachine.withgoogle.com/train>

≡ Teachable Machine

Class 1

41 Image Samples

Webcam Upload

Class 2

30 Image Samples

Webcam Upload

Add a class

Training

Model Trained

Advanced

Epochs: 50

Batch Size: 16

Learning Rate:
0.001

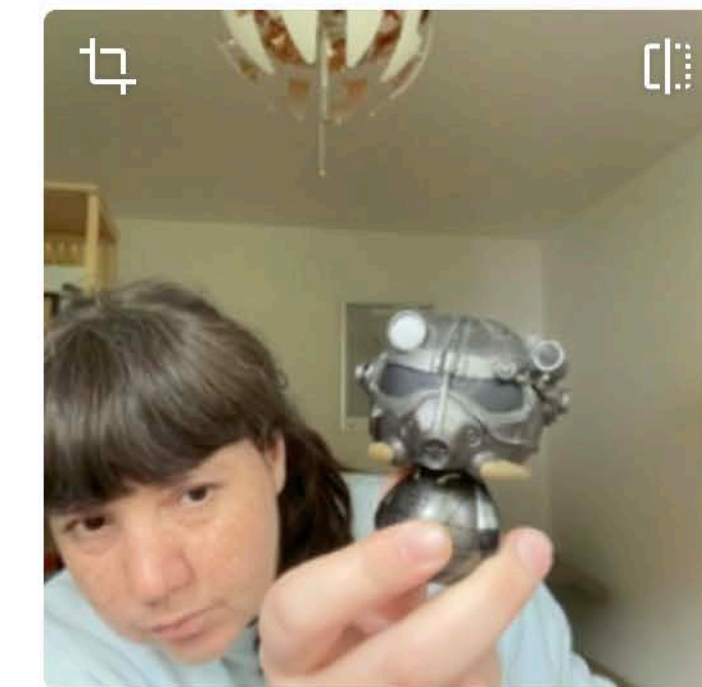
Reset Defaults

Under the hood

Preview

Export Model

Input ☐ ON Webcam



Output



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Practical Task

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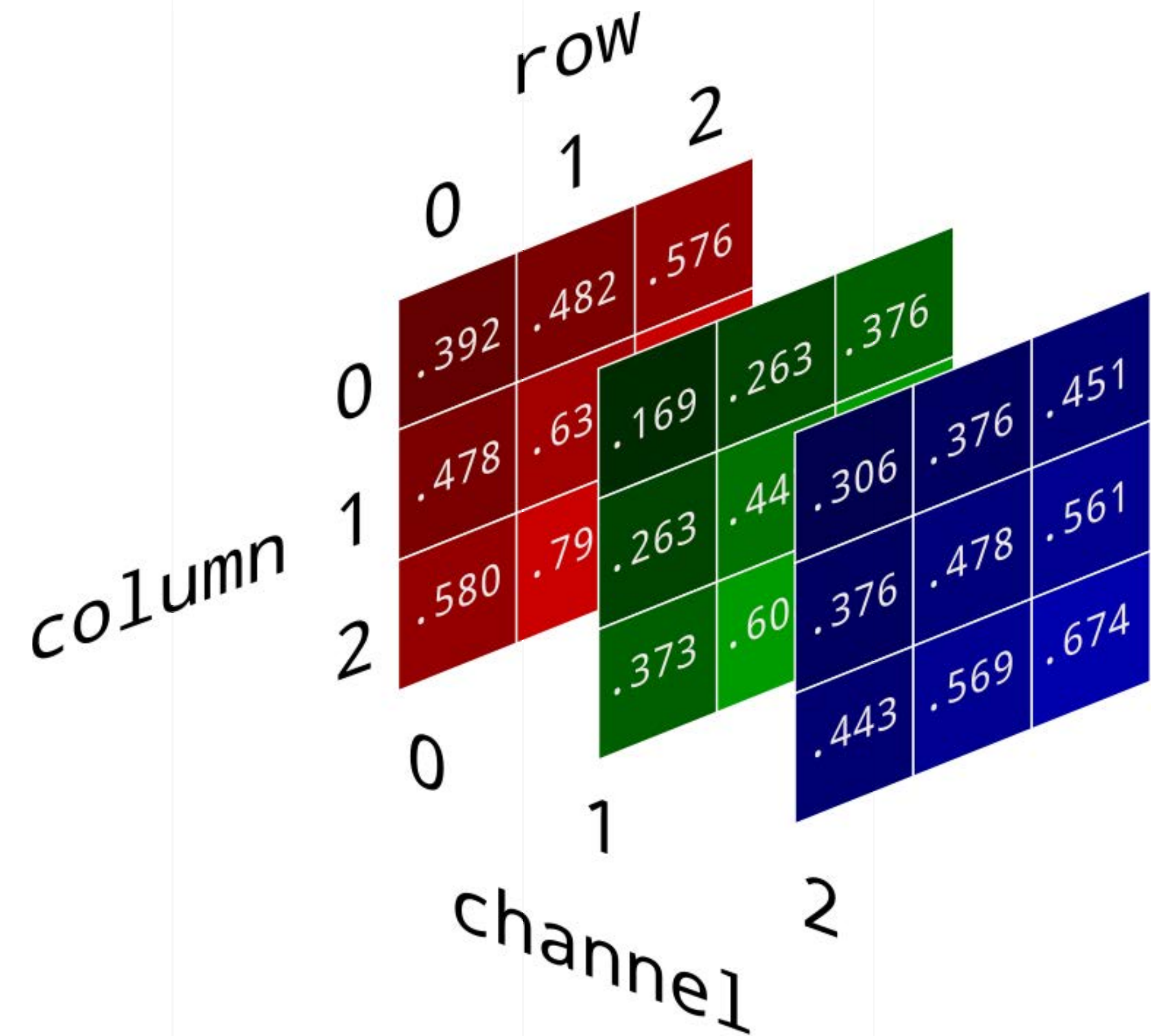




An image is an array or matrix, consisting of 3 channels and rows + columns

For greatest flexibility we can pass Images to numpy array to do scientific operations on it.

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Load an Image

```
img = cv2.imread(filename)
```

Display the Image

```
cv2.imshow('image', img)
```

Save the Image

```
cv2.imwrite(new_filename,  
image)
```





Basic Image Processing: learn how to load, display, and save an image using Python and OpenCV.

- Import the necessary libraries: Python's cv2 for OpenCV functions and matplotlib for displaying the image.
- Load an image using `cv2.imread()`.
- Display the image using matplotlib or OpenCV
- Save the image to a specified location using `cv2.imwrite()`





Kick Off Questionnaire

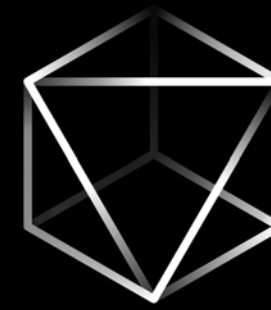
Thanks & Leave Feedback



[https://
forms.microsoft.com/e/
5Hv1wb98Su](https://forms.microsoft.com/e/5Hv1wb98Su)

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AslihanYilmaz-Hauck|AI&ComputerVisionWS25

HSDDüsseldorf|BMI